Exact Structural Abstraction and Tractability Limits

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Abstract

Any rigorously specified problem determines an admissible-output relation R, and exact correctness depends only on the induced classes $s \sim_R s' \iff \mathrm{Adm}_R(s) = \mathrm{Adm}_R(s')$. Exact relevance certification asks which coordinates recover those classes. Decision, search, approximation, statistical, randomized, horizon, and distributional guarantees all reduce to this same quotient-recovery problem. Tractable cases still admit a finite primitive basis, but optimizer-quotient realizability is maximal, so quotient shape alone cannot mark the frontier.

For frontier theorems, orbit gaps are the exact obstruction. Exact classification by closure-law-invariant predicates succeeds exactly when the target is constant on closure orbits; on a closure-closed domain, this is equivalent to disjointness of the positive and negative orbit hulls, and when it holds there is a least exact closure-invariant classifier. Across four natural candidate tractability predicates, a uniform pair-targeted affine witness produces same-orbit disagreements and rules out exact structural classification on the full binary pairwise domain. Because the canonical optimizer-set exact specifications of that witness class are already rigorously specified problems, no universal exact-certification characterization over formal problems escapes the same obstruction; this is by internal witness class, not by asserting that every problem is binary pairwise. Restricting the domain helps only by removing orbit gaps. Approximation also has a strict limit: without explicit gap control, arbitrarily small perturbations can flip relevance and sufficiency.

1 Introduction

Any rigorously specified computational problem already determines a state-indexed admissible-output relation R, and the only state distinctions that matter are the admissible-output equivalence classes $s \sim_R s' \iff \mathrm{Adm}_R(s) = \mathrm{Adm}_R(s')$.

Informally: knowing what matters is the only thing that matters.

Every exact correctness claim reduces to the same quotient-recovery problem (Theorem 2.37). Here exact means exact agreement with the admissible-output relation itself, not zero-error determinism or the absence of approximation, randomization, statistical thresholds, or failure states inside the specification. Section 2 proves this universality explicitly: arbitrary exact output semantics transfer to exact relevance certification (Corollary 2.33); approximation, PAC/regret/risk, and randomized-output guarantees are named instances of the same reduction (Corollaries 2.48, 2.44, and 2.45); every exact claim admits a coordinate presentation (Proposition 2.40); and every state equivalence relation is realizable at this semantic level (Corollary 2.54). The construction is direct. Its force is conceptual: one quotient semantics already covers these exact output formalisms. The optimizer-quotient framework of [25] isolates this canonical semantic layer, and the earlier optimizer-quotient factorization results are recovered as exact-semantics special cases of the more general admissible-output quotient theorem.

The validity relation is the correctness condition itself, not an optional encoding trick. Different coordinate presentations may make quotient recovery trivial or rich, but they do not change the canonical quotient itself. For example, if a specification declares every sorted output admissible for a given input list, then two states are equivalent exactly when they admit the same sorted outputs, and relevance asks which coordinates recover that admissible-output class. If a specification instead declares every hypothesis with error at most ε admissible, the same question asks which coordinates recover that admissible hypothesis class. Proposition 2.40 gives the universal one-coordinate realization, while Propositions 2.41–2.43 give structurally informative realizations of the same semantics. SAT fits through its exact yes/no validity predicate, sorting through the set of correctly sorted outputs for each input, and fixed approximation or PAC guarantees through the outputs meeting the stated threshold. The frontier theorems therefore concern tractable recovery of a canonical quotient across presentations, not existence of semantics.

The paper has two threads. Section 2 establishes the universal semantic reduction from exact admissible-output specifications to quotient recovery (Theorem 2.37 and Corollary 2.38). Sections 3–6 ask whether that universal object admits a finite structural frontier theorem. The two threads meet in Sections 2 and 6, where theorem-forced closure invariance turns the semantic quotient framework both into a no-go for finite local frontier classifiers and into a positive orbit algebra of exact classification: once exact semantics are fixed, tractability classification must respect the presentation equivalences that leave the underlying exact-certification problem unchanged (Proposition 2.18 and Theorem 2.20). Universality therefore strengthens the no-go rather than weakening it: Corollary 6.17 shows that no universal exact-certification treatment over rigorously specified problems escapes the obstruction, because the canonical optimizer-set exact specifications of the full binary pairwise witness domain are rigorously specified problems already. The point is not that every individual problem is binary pairwise. The point is that any universal treatment must already restrict correctly to that witness class.

The orbit-gap contradiction depends only on closure-law invariance (Theorem 6.11), and orbit gaps are the complete obstruction criterion for exact classification by closure-law-invariant predicates (Proposition 6.18). The same framework also has a significant positive side: closure-invariant predicates are exactly the fixed points of orbit saturation, exact classification on a closure-closed domain is exactly hull disjointness, and whenever exact classification is possible there is a least exact closure-invariant classifier (Proposition 6.20, Theorem 6.21, and Corollary 6.22). Closure-law invariance is itself forced by correctness (Theorem 2.20). Closure-equivalent representations encode the same certification problem (Proposition 2.18), so a correct tractability classifier must assign them the same verdict. The verdict concerns tractability of the underlying problem, not tractability of a specification format. Representative compute-cost instantiations include optimizer computation, canonical payload and search tasks, and external or transported outputs with explicit admissibility-preserving transport; the same orbit-gap mechanism therefore applies there as well as to feature sufficiency. The four named obstruction families witness the generic orbit-gap criterion for several representation-level predicates. Corollary 6.14 and Corollary 6.16 instead state the finite-structural inaccessibility result for tractability: the abstract closure-hull classifier exists, but it does not belong to the finite structural class on the stated domain. Corollary 6.23 gives the exact domain-relative statement: restricting the domain helps precisely when the restricted closure-closed domain contains no orbit gaps for the target notion. Definition 2.19 specifies the finite structural class used below; the contradiction itself uses only closure-law invariance.

Once correctness forces closure-orbit agreement, no finite structural classifier can separate the obstruction families (Corollary 6.14).

A natural first frontier attempt is to classify tractable families by local representation patterns: bounded treewidth, symmetry signatures, supported interaction graphs, and other finite motifs extracted directly from the utility presentation. Definition 2.19 isolates that direct structural-classification regime: efficiently checkable, presentation-independent, structurally extractable verdicts whose definitions are built from bounded local patterns rather than unbounded global computation. Closure-invariance is forced by correctness (Theorem 2.20); polynomial-time checkability, structural extractability, and bounded-pattern definability act only as guardrails restricting attention to direct representation-level structure.

On the positive side, we first show that all known tractable cases collapse to eight primitive mechanisms, together with regime lifts and degenerate collapses (Corollary 3.3). Bounded distinct action profiles then compress to bounded-action slices without changing exact certification, extending the bounded-actions theorem by an explicit reduction (Proposition 2.26). The eight-mechanism basis is the inventory of the positive landscape developed in the paper, not a complete classification of all tractability mechanisms for exact relevance certification. For practitioners, this gives a usable message: these eight mechanisms explain the tractable cases currently on the table, but the later no-go shows that this finite local framework cannot be turned into a complete automatic frontier test. Optimizer realizability is maximal (Theorem 5.1), so quotient shape alone is too expressive to classify tractability; the same theorem also gives representational universality, because arbitrary label kernels already embed as optimizer quotients. The point is not that the realizing construction is deep, but that no quotient-level obstruction remains in the unrestricted regime.

Exactness is not arbitrary. At zero distortion, any summary that preserves optimal actions must still separate distinct optimizer classes: if two different optimizer classes were assigned the same summary value, the summary would merge states with different optimal-action sets and would cease to be lossless (Proposition 2.8). Exact relevance therefore marks the theoretical floor for lossless abstraction. Section 2 also identifies the approximation boundary: approximate surrogate claims about relevance or sufficiency are admissible only with an explicit stability reduction to the exact optimizer sets, because witness-gap control preserves the corresponding witness (Proposition 2.49), a uniform strict-gap hypothesis preserves the entire optimizer quotient together with the sufficient, relevant, and minimal-sufficient structure (Proposition 2.52), and arbitrarily small uniform perturbations can otherwise flip the exact judgment (Corollary 2.51). In applied settings, this is the practical warning: approximation alone does not justify claims about which coordinates matter.

Realizability of arbitrary quotients fixes the organizing principle: tractability predicates must be evaluated on representation-level structural classes, but modulo theorem-forced equivalences of presentation (Theorem 5.1). The framework is therefore universal representationally as well as semantically: arbitrary exact semantics reduce to quotient recovery, and arbitrary label kernels already occur as optimizer quotients. In particular, closure under relabelings, positive affine utility reparameterizations, duplication, and binary irrelevant-coordinate extension is not a modeling preference; it is inherited from the fact that exact certification depends only on the induced decision quotient relation (Proposition 2.18).

The closure laws used below are an explicit theorem-verified generating family of presentation moves:

- action and coordinate relabeling,
- statewise positive affine utility reparameterization,
- action duplication and state duplication,
- binary irrelevant-coordinate extension.

Each preserves the underlying exact-certification problem up to explicit transport (Proposition 2.18). The list is not claimed maximal. Any larger quotient-preserving equivalence only strengthens the impossibility results.

A closure-closed domain is a family of representations closed under those theorem-forced presentation moves. Quotient-respecting representation classes are naturally closure-closed. Families defined by a fixed syntactic normal form or by presentation-sensitive side conditions often are not.

Standing Semantics

For a decision problem with state space S, action space A, and utility U, write

$$\mathrm{Opt}(s) = \{a \in A : U(a, s) \text{ is maximal among all actions at } s\}.$$

This induces the decision-equivalence relation

$$s \sim_{\mathrm{Opt}} s' \iff \mathrm{Opt}(s) = \mathrm{Opt}(s').$$

The associated optimizer quotient is the quotient set $S/{\sim_{\mathrm{Opt}}}$. A coordinate set I is *sufficient* when agreement on I forces equality of optimizer classes, and a coordinate is *relevant* when deleting it destroys sufficiency. Later closure laws and obstruction families are all phrased relative to this quotient viewpoint.

For a concrete two-action, two-coordinate example, let the state space be $\{0, 1\}^2$ with coordinates (x_0, x_1), let the actions be a, b, and define

$$U(a, x) = x_0, \qquad U(b, x) = 0.$$

Then

$$\mathrm{Opt}(00) = \mathrm{Opt}(01) = \{a, b\}, \qquad \mathrm{Opt}(10) = \mathrm{Opt}(11) = \{a\}.$$

The optimizer quotient therefore has exactly two classes, namely $\{00, 01\}$ and $\{10, 11\}$. The coordinate set $\{0\}$ is sufficient, coordinate 1 is irrelevant, and coordinate 0 is relevant. Only the first coordinate changes what matters.

Unrestricted predicates trivialize any impossibility theorem via direct semantic encoding. A finite structural class is therefore needed: polynomial-time checkability, structural extractability, closure-law invariance, and bounded-pattern definability. This class excludes direct semantic encoding while still covering the structural results established here. Closure-law invariance is the clause used in the contradiction (Theorem 6.11); the remaining clauses restrict the search space to finite, explicitly structural predicates.

The theorem is a no-go for classifiers inside the finite structural class, not a no-go for tractable classes themselves. The regime is deliberately scoped: it is strong enough to exclude direct semantic encoding, yet narrow enough that membership is still an explicit representation-level condition. Stronger representation-sensitive structure lies outside Definition 2.19; the algebraic invariants used in CSP dichotomy theorems are not bounded-pattern definable under this locality restriction. The result is therefore compatible with CSP-style positive frontier theorems. The open problem is to identify the successor structural class using stronger structure of that kind.

Within this finite structural class, the no-go template is witnessed by same-orbit pairs with different obstruction status. The key mechanism is an action-independent, pair-targeted additive affine term (the statewise "α-component" of positive affine transport), which changes the target predicate while preserving closure equivalence. Closure-law invariance then transports any candidate predicate across the orbit and forces contradiction.

The dominant-pair family is the canonical instance, and the same mechanism applies to margin masking, ghost-action concentration, and additive/statewise offset concentration. A dedicated worked dominant-pair orbit example appears in Section 6. The force of using all four families is robustness across multiple natural candidate tractability predicates, not four unrelated proof devices. Theorem 6.11 identifies closure-law invariance as the hypothesis used in the contradiction, and Theorem 2.20 upgrades the scope from invariant classifiers to correct classifiers on closure-closed domains. The remaining clauses delimit the direct finite structural regime. The open problem is to identify the stronger structural principle that yields a correct frontier theorem.

Notation Summary

Notation	Meaning
$\mathrm{Adm}_R(s)$	admissible outputs at state s under exact specification R
$s \sim_R s'$	admissible-output equivalence: $\mathrm{Adm}_R(s) = \mathrm{Adm}_R(s')$
$S/{\sim_R}$	quotient of states by admissible-output equivalence
$\mathrm{Opt}(s)$	optimizer set of a decision problem at state s
$s \sim_{\mathrm{Opt}} s'$	optimizer-quotient equivalence: $\mathrm{Opt}(s) = \mathrm{Opt}(s')$
I sufficient	agreement on coordinates I forces equality of quotient classes
i relevant	erasing coordinate i destroys sufficiency
closure orbit	all representations connected by theorem-forced closure operations
slice	a concrete binary pairwise utility presentation used by the frontier program

An accompanying Lean development mechanically verifies the realizability constructions, the generic orbit-gap argument, and the family-level admissible no-go theorems used in the collapse argument; the archived artifact is available at `https://doi.org/10.5281/zenodo.19457896`.

2 Exact Semantics and Structural Preconditions

Realizability of arbitrary quotients leaves a more precise frontier question. If quotient shape alone is too expressive, what should replace it as the organizing unit of the metatheorem?

At minimum, the tractability frontier should not classify arbitrary sets of decision problems. It should range over representation-level structural classes and it should ignore benign changes of presentation.

The right eventual condition list will almost certainly be stronger than this minimal principle. For example, a mature frontier theorem may also require closure under adding explicitly irrelevant coordinates, adjoining dummy actions, or other representation-preserving refinements. But action/state relabel invariance is the irreducible starting point.

Proposition 2.1[1] (Exact Certification Depends Only on the Decision Quotient Relation). *If two decision problems on the same state space induce the same decision-equivalence relation, then they induce the same exact-certification structure. In particular, they have the same sufficient coordinate sets, the same minimal sufficient coordinate sets, and the same relevant and irrelevant coordinates.*

[1] Lean: `FR30-35`, `FR37`, `FR51`

Moreover, their quotient objects are canonically equivalent; in finite settings, those quotient objects therefore have the same cardinality.

The decision quotient relation determines exact certification. Equality of optimizer maps is only one sufficient route to that quotient data, because equal optimizer maps induce the same decision-equivalence relation.

Corollary 2.2[2] (Sufficiency Is Relation Refinement). *A coordinate set I is sufficient if and only if the coordinate-agreement relation induced by I refines the decision quotient relation.*

Sufficiency is purely relational: the coordinate-agreement relation induced by I must refine the decision quotient relation. Utility language drops out entirely.

Corollary 2.3[3] (Relevance Is Failure of Erased Sufficiency). *A coordinate i is irrelevant if and only if deleting it leaves a sufficient coordinate set, namely* $\mathrm{univ} \setminus \{i\}$*. Equivalently, i is relevant if and only if that erased set is not sufficient.*

Corollary 2.3 makes the role of individual coordinates canonical: relevance is exactly the obstruction to retaining sufficiency after coordinate erasure.

The underlying definition is already witness-based: a coordinate is relevant exactly when there exist two states that agree on all other coordinates but have different optimizer classes.

Corollary 2.4[4] (Certification Statistics Factor Through the Quotient Relation). *Any statistic that is a function of the sufficient-set family and relevant-coordinate set is invariant under equality of the decision quotient relation. In particular, the number of sufficient coordinate sets, the number of minimal sufficient coordinate sets, and the number of relevant coordinates are quotient invariants.*

Every certification statistic built from sufficient sets or relevant coordinates factors through the decision quotient relation. Counts of sufficient sets, minimal sufficient sets, and relevant coordinates are therefore quotient invariants.

Proposition 2.5[5] (Minimal Sufficient Sets Are Canonical on Product Spaces). *Assume the coordinate space is a product space with decidable coordinate equality, and let I be a minimal sufficient set. Then for every coordinate i,*

$$i \in I \iff i \text{ is relevant.}$$

Consequently, whenever a minimal sufficient set exists, it is unique.

Proof sketch. Minimality forces every coordinate in I to matter, because otherwise erasing it would preserve sufficiency and contradict minimality. Conversely, every relevant coordinate must lie in every sufficient set, hence in I. The displayed equivalence therefore identifies I with the relevant-coordinate set, and uniqueness follows immediately. ■

Corollary 2.6[6] (Sufficient Sets Form a Principal Filter). *Under the same product-space assumptions, if I is a minimal sufficient set, then a coordinate set J is sufficient if and only if $I \subseteq J$.*

Proof sketch. By Proposition 2.5, the minimal sufficient set is exactly the relevant-coordinate set. Every sufficient set must contain all relevant coordinates, and upward closure of sufficiency gives the converse. ■

These two statements give a genuinely positive global structure theorem: once a minimal sufficient set exists, the sufficient-set family is not arbitrary. It is completely organized by one canonical generating set.

[2] : `FR48` [3] : `FR49-50` [4] : `FR41-44` [5] : `FR352` [6] : `FR353`

Corollary 2.7[7] (Structural Rank Lower-Bounds Any Sufficient Description). *Let $\mathcal{D}$ be a decision problem with structural rank* $\text{srank}(\mathcal{D})$*, and let I be any sufficient coordinate set. Then*

$$\text{srank}(\mathcal{D}) \le |I|.$$

In particular, no sufficient description can use fewer coordinates than the number of relevant coordinates.

Proof sketch. Every sufficient set contains all relevant coordinates, and structural rank is exactly the cardinality of the relevant-coordinate support. The inequality follows immediately. ■

Proposition 2.8[8] (Zero-Distortion Summaries Refine the Optimizer Quotient). *Let $\sigma : S \to C$ be a summary map. Suppose that*

$$\sigma(s) = \sigma(s') \implies Opt(s) = Opt(s') \qquad \textit{for all } s, s' \in S.$$

Then each summary fiber is contained in a single optimizer-quotient class. Equivalently, σ factors through the optimizer quotient relation, and distinct optimizer classes require distinct summary symbols.

Proof. If $\sigma(s) = \sigma(s')$, the hypothesis gives $\text{Opt}(s) = \text{Opt}(s')$. So every σ-fiber is contained in one decision-equivalence class. This is exactly the statement that σ refines the optimizer quotient. Conversely, if two distinct optimizer classes shared a summary symbol, choosing representatives from those classes would violate the zero-distortion premise. ■

Proposition 2.8 explains why exact relevance is not merely one point on an approximation spectrum. Once distortion is required to be zero, lossless decision-preserving abstraction is already constrained by the optimizer quotient itself. Later propositions in this section sharpen the approximation boundary: approximate surrogate claims about relevance need explicit stability control relative to the exact optimizer sets, because uniform closeness alone does not preserve the exact decision boundary.

Corollary 2.9[9] (Constant-Optimizer Collapse Forces Trivial Certification). *If the optimizer set is constant across all states, then the empty coordinate set is sufficient and every coordinate is irrelevant.*

Corollary 2.9 isolates the first genuinely degenerate mechanism in the tractable basis. The certification problem disappears because there is no state dependence left to certify.

Proposition 2.10[10] (Explicit Enumeration Is Parameter-Dependent, Not Structural). *For every fixed exponent k, there exists a Boolean-cube family with state space size $2^n > n^k$. Thus explicit-state enumeration can outrun any fixed monomial bound in the ambient dimension parameter.*

Brute-force state enumeration depends on a separate size parameter and does not arise from an optimizer-compatible mechanism like symmetry, low rank, or tree structure.

Proposition 2.11[11] (Positive Affine Utility Reparameterizations Are Invisible). *Statewise positive affine transformations of utility leave exact certification unchanged. In particular, replacing*

$$U(a,s) \qquad \textit{by} \qquad \alpha(s) + \beta(s)U(a,s)$$

with $\beta(s) > 0$ for every state s does not change sufficient coordinate sets or relevance judgments.

[7] : `FR354` [8] : `FR176-177` [9] : `FR52-53` [10] : `FR66-67` [11] : `FR13-16`

Statewise positive affine utility reparameterization is invisible to exact certification: only the induced decision quotient matters, not the particular positive affine scale used to encode utility.

Proposition 2.12[12] (Duplicate Actions and Duplicate States Preserve Certification). *Duplicating a single action without changing its utility profile preserves the decision quotient relation and hence preserves sufficiency and relevance. At the optimizer-set level, the only possible change is the addition of the duplicate action itself: original optimal actions remain optimal, and the duplicate is optimal exactly when the original action was. Duplicating a single state without changing its utility profile preserves the decision quotient up to the obvious quotient equivalence and also preserves sufficiency and relevance.*

Proposition 2.12 gives two more theorem-forced closure laws. These are not modeling conventions; they are consequences of the fact that exact certification factors through quotient data rather than through accidental multiplicity in the presentation.

Proposition 2.13[13] (Relabeling Invariance Is Forced by Exact Certification). *Action relabeling preserves optimizer equivalence and exact sufficiency. State relabeling does so as well once the coordinate structure is transported along the state bijection. Consequently, any structural tractability frontier for exact relevance certification must be closed under these relabelings.*

Bijective relabelings do not change the optimizer map up to the obvious identification, so they cannot change exact certification.

The second necessary ingredient is a genuine expressivity restriction. Call a class *kernel-universal* if, for every labeling map $\phi : S \to T$, it contains some decision problem whose optimizer quotient realizes exactly the kernel partition of ϕ. By Theorem 5.1, kernel universality is an extremely strong property.

The canonical finite invariant here is the size of the optimizer quotient, equivalently the number of distinct optimal-action sets. The quotient is identified with the range of the optimizer map, so quotient size is not auxiliary bookkeeping; it is intrinsic.

Proposition 2.14[14] (Quotient Size Is Unbounded Under Realizability). *For every $m \in \mathbb{N}$, there exists a finite decision problem whose optimizer quotient has size exactly m.*

Proof. Take $m = k + 1$ and realize the identity labeling on $\mathrm{Fin}(k+1)$. Proposition 5.3 identifies the quotient object with the range of that identity labeling, which has cardinality $k + 1$. Equivalently, Proposition 5.2 realizes the discrete equivalence relation on $\mathrm{Fin}(k+1)$. ∎

A meaningful frontier statement cannot range over classes that are simultaneously relabel-invariant and kernel-universal, because quotient shape alone then becomes too expressive to isolate the boundary.

Proposition 2.15[15] (Invariance Alone Is Too Weak). *The universal class of all decision problems satisfies action relabeling invariance and state relabeling invariance, but it is kernel-universal. Consequently, relabeling invariance by itself is far too weak to isolate a tractability frontier.*

Benign invariance axioms and expressivity-limiting axioms play different roles in a frontier theorem.

1. **Benign invariance axioms** say the family should ignore arbitrary naming choices.

2. **Expressivity-limiting axioms** say the family should not realize arbitrary quotient structure wholesale.

[12] : `FR54-56`, `FR68-73` [13] : `FR17-20` [14] : `FR24`, `FR38` [15] : `FR21-23`

Any genuine optimizer-compatible frontier theorem will need both. Without decision-quotient invariance and its immediate corollaries such as relabel invariance and positive affine invariance, the statement is not structural. Without an expressivity restriction excluding kernel universality or something comparably strong, the realizability theorem and Proposition 2.14 make the class too large for quotient shape alone to carry the boundary.

Proposition 2.16[16] (Independent Summary Frameworks Converge on the Same Structural Core). *Write $r = \mathrm{srank}(\mathcal{D})$ for the number of relevant coordinates of a Boolean decision problem, and let m be the number of optimizer-quotient classes. Then:*

1. *$m \le 2^r$, so the counting entropy $\log_2 m$ is at most r;*
2. *the diagonal $0/1$ relevance-support matrix has rank exactly r;*
3. *every zero-distortion decision-preserving summary requires at least m distinct summary symbols.*

Hence quotient entropy, Fisher-style support counting, and zero-distortion summary size are all controlled by the same support/quotient core.

Proof. If two states agree on all relevant coordinates, then they differ only on irrelevant ones. Changing those irrelevant coordinates one at a time cannot change the optimizer, so the relevant coordinates already determine the optimizer class. Therefore there are at most 2^r optimizer classes, one for each Boolean assignment on the relevant support, and $\log_2 m \le r$ follows.

For the second clause, the diagonal relevance-support matrix has a 1 exactly on relevant coordinates and a 0 elsewhere. Its rank is therefore the number of relevant coordinates, namely r.

The third clause is exactly Proposition 2.8: a zero-distortion summary must assign distinct symbols to distinct optimizer classes. ■

Proposition 2.16 does not identify entropy, support counting, and lossless summaries as identical notions. It shows that they already converge on the same structural content before any admissibility axiom is imposed. Closure-law invariance follows from this convergence: once several independent frameworks ignore surface presentation in favor of the support/quotient core, a structural tractability classifier should do the same. The remaining admissibility clauses serve as algorithmic and semantic guardrails.

From Necessary Conditions to a Finite Structural Class

The closure properties above are theorem-forced by exact-certification semantics, but they are not enough on their own to support a meaningful impossibility theorem. If one allows unrestricted predicates, a semantic predicate can simply encode the target complexity class and trivialize the statement.

Proposition 2.17[17] (Unrestricted Predicates Trivialize Exact Characterization). *Without admissibility restrictions, there exist normalization predicates whose membership condition directly encodes the intended polynomial-time side of the landscape, so unrestricted collapse-impossibility claims fail.*

Proposition 2.17 is not a by-product of the proof method. It is the expected behavior of unrestricted semantic predicates. Any meaningful frontier theorem must therefore be parameterized by a structural class that simultaneously (i) respects theorem-forced invariances and (ii) blocks direct semantic encoding.

[16] : `FR178-182` [17] : `FR145`

A natural first attempt is to classify tractable families by local structural signatures visible in the presentation itself: bounded treewidth, symmetry traces, support graphs, or other finite graph-pattern conditions extracted from binary pairwise syntax. The finite structural class below formalizes that first attempt under theorem-forced invariance.

Proposition 2.18[18] (Closure Operations Preserve Exact Certification). *The theorem-forced closure operations preserve the exact-certification problem itself. More precisely:*

1. *action relabeling, coordinate relabeling, and statewise positive affine reparameterization preserve sufficient coordinate sets and relevant coordinates under the evident transport;*
2. *action duplication and state duplication preserve sufficient coordinate sets and relevant coordinates exactly;*
3. *binary irrelevant-coordinate extension preserves sufficiency after lifting coordinate sets, preserves relevance on the original coordinates, and makes the new coordinate irrelevant.*

In particular, each closure step induces the same exact-certification decision problem up to an explicit coordinate-set transport. For the irrelevant-coordinate case in the present binary-pairwise formalization, the encoding blowup is only constant-factor because the added coordinate is binary.

Proof sketch. Each clause is verified directly at the level of exact-certification semantics. The relabeling and positive-affine cases preserve the decision-equivalence relation, hence preserve sufficiency and relevance. The duplication cases preserve decision-equivalence through the explicit quotient projections back to the original problem. The irrelevant-coordinate case is binary noise extension: $I \subseteq [d]$ is sufficient for the base slice if and only if its lift is sufficient after extension, the original coordinates remain exactly the relevant ones, and the new coordinate is irrelevant. Table 1 summarizes the resulting transport statement. The encoding remarks in the right-hand column are direct from the concrete binary-pairwise representation and are not separate theorem claims. ■

Operation	**Exact-certification transport**	**Encoding effect**
Action/state relabeling	same sufficient sets and relevant coordinates after transport	relabeling only
Positive affine reparameterization	same sufficient sets and relevant coordinates	same arity, same action set; utility magnitudes rescaled
Action/state duplication	same sufficient sets and relevant coordinates	carrier duplication only
Binary irrelevant-coordinate extension	$I \leftrightarrow \mathrm{lift}(I)$, old relevance preserved, new coordinate irrelevant	arity increases by one binary coordinate

Table 1: Closure-operation transport for exact certification. The transport column records the exact-certification invariance stated in the closure theorems; the encoding-effect column records the direct representation-level effect of each operation in the binary-pairwise formalization.

Definition 2.19 (Finite Structural Predicate). A normalization predicate belongs to the finite structural class when it satisfies all of the following:

[18] : `FR15-20`, `FR55-60`, `FR83-85`

1. polynomial-time checkability at the slice level;
2. invariance under the closure laws forced by exact certification (relabelings, positive affine reparameterizations, duplication operations, and binary irrelevant-coordinate extension);
3. structural extractability of the associated dependency graph;
4. bounded-pattern definability.

More explicitly, let $|U|$ denote the encoding size of a binary pairwise slice U, and let $X(U)$ be its canonical finite pairwise syntax. Polynomial-time checkability means that there exist constants $c, k \in \mathbb{N}$ and a decision procedure A such that

$$A(U) = 1 \iff Q(U), \qquad \mathrm{time}_A(U) \leq c(|U| + 1)^k + c.$$

Structural extractability means that whenever $Q(U)$ holds, the associated coordinate graph is obtained from syntax alone: there exists a uniform extractor

$$E : X \mapsto G_E(X)$$

on finite pairwise syntactic presentations such that the graph attached to U is exactly $G_E(X(U))$.

Bounded-pattern definability is likewise finite and explicit. Concretely, there must exist integers

$$r_{\max},\ n_{\max},\ a_{\max},\ c_{\max} \in \mathbb{N}$$

and finite sets $\mathcal{W}, \mathcal{F}$ of rooted local patterns such that every pattern in $\mathcal{W} \cup \mathcal{F}$ has radius at most $r_{\max}$, uses at most $n_{\max}$ vertices, involves at most $a_{\max}$ action labels, and all listed coefficient magnitudes are at most $c_{\max}$. Write

$$\mathcal{N}_U(v) := \mathcal{N}_{r_{\max}}(X(U), v)$$

for the rooted radius-$r_{\max}$ neighborhood of a vertex v in the extracted finite syntax $X(U)$, and write $P \sqsubseteq \mathcal{N}_U(v)$ when the rooted pattern P occurs in that neighborhood. Then membership in Q is determined by these finite patterns alone:

$$Q(U) \iff \Big(\exists v\ \exists W \in \mathcal{W} :\ W \sqsubseteq \mathcal{N}_U(v)\Big)\ \wedge\ \Big(\forall v\ \forall F \in \mathcal{F} :\ F \not\sqsubseteq \mathcal{N}_U(v)\Big).$$

Thus the definition ranges over a fixed finite vocabulary of local rooted configurations; it does not call an unbounded global computation under a structural name.

Polynomial-time checkability is the minimal algorithmic requirement. A tractability characterization that cannot itself be recognized efficiently only relocates the computational difficulty from exact certification to the membership test for the classifier.

Closure-law invariance is the semantic requirement used in the orbit-witness arguments of Section 6. It is forced by the Block 6 invariance theorems, because exact certification itself is unchanged by relabelings, positive affine utility reparameterizations, duplication operations, and binary irrelevant-coordinate extensions. These operations are an explicit theorem-verified generating family of quotient-preserving presentation moves. They are not claimed maximal.

Structural extractability distinguishes structural characterizations from purely semantic ones. Without such a condition, one can define predicates directly on optimizer behavior or on the solved certification instance itself, bypassing the stated aim of classifying tractability by representation-level structure.

Bounded-pattern definability is the locality restriction. Its role is to formalize the local-pattern heuristic directly and to exclude predicates whose membership test is implemented by an arbitrarily large finite schema or other hidden global computation. The contradiction does not use this clause. The clause restricts attention to the direct finite local regime.

The compute-cost layer adds no new orbit mechanism. Once correctness forces closure-orbit agreement for one exact task, each output-production variant inherits the same conclusion through its explicit transport law. The next results record representative instantiations.

Theorem 2.20[19] (Tractability Classifiers Are Forced to Be Closure-Invariant). *Let C be any procedure assigning to each representation in a closure-closed domain Γ a verdict in* $\{\text{tractable}, \text{intractable}\}$*, and suppose that C correctly predicts whether exact certification is polynomial-time decidable on the underlying problem. Then C must agree on representations within the same closure orbit. Equivalently, every correct tractability classifier on a closure-closed domain is closure-invariant on that domain, regardless of whether its internal features are themselves invariant.*

Proof sketch. If $U, V \in \Gamma$ lie in the same closure orbit, Proposition 2.18 shows that they induce the same exact-certification problem up to the explicit coordinate-set transport attached to each closure step. The theorem uses the semantic notion of correctness stated above: C is correct exactly when its verdict matches whether the underlying exact-certification problem is polynomial-time decidable. For a single closure step, the transported coordinate-set instance has the same yes/no answer as the original one, so a polynomial-time decision procedure for one representation yields a polynomial-time decision procedure for the other with only the polynomial overhead of the transport. Hence tractability is unchanged by each closure step. A closure orbit is generated by composing such steps, so tractability is constant on the orbit, and any correct classifier must return the same verdict on U and V. ■

Informally: a correct classifier gets one answer for one problem.

Theorem 2.21[20] (Compute-Cost Version: Optimizer Computation). *Let C be any procedure assigning to each representation in a closure-closed domain Γ a verdict in* $\{\text{tractable}, \text{intractable}\}$*, and suppose that C correctly predicts whether optimizer computation is polynomial-time solvable on the underlying problem. Then C must agree on representations within the same closure orbit.*

Proof sketch. The optimizer-computation case is another instance of the same correctness-on-domain theorem. Each closure step preserves polynomial-time solvability of optimizer computation under its explicit state and output transports. Correctness of C therefore forces the same verdict on closure-equivalent representations. ■

Corollary 2.22[21] (Compute-Cost Version: Canonical Payload and Search Tasks). *The same closure-orbit agreement conclusion holds when C correctly predicts polynomial-time solvability of deterministic optimizer-set payload output. It also holds when C correctly predicts polynomial-time solvability of admissible-output search on optimizer-set semantics.*

Proof sketch. Apply the same correctness-on-domain theorem to the two canonical compute-cost families built from optimizer-set payload output and optimizer-set admissible-output search. ■

These are representative instances of one pattern rather than new obstruction mechanisms. The same correctness-forces-invariance theorem therefore covers the compute-cost layer as well.[22]

[19] : `FR197-199`, `FR245-249` [20] : `FR248` [21] : `FR259-262` [22] : `FR245-249`, `FR259-262`

Corollary 2.23[23] (Compute-Cost Version: External Output Objects). *Let X be an external output class whose admissibility relation is preserved under the closure laws by pullback on states and identity on outputs. Then any correct classifier for polynomial-time search over admissible X-outputs on a closure-closed domain must agree on closure orbits, and the same orbit-gap no-go applies.*

Proof sketch. The external-output case transports only the input state; the output object itself is unchanged. Once the admissibility relation respects those state pullbacks, the same correctness-on-domain theorem applies. ■

Corollary 2.24[24] (Compute-Cost Version: Representation-Relative Output Objects). *Let X_U be a representation-relative output object whose type and admissibility relation may vary with the representation U, and suppose each closure witness carries an explicit output transport preserving admissibility. Then any correct classifier for polynomial-time search over admissible X_U-outputs on a closure-closed domain must agree on closure orbits, and the same orbit-gap no-go applies. Named instances include representation-relative hypotheses, estimators, policies, and randomized procedures.*

Proof sketch. The transported-output case packages the witness-by-witness output maps together with the corresponding admissibility preservation laws. The same correctness-on-domain theorem then applies to representation-relative hypotheses, estimators, policies, and randomized procedures. ■

Correctness already forces the same orbit agreement. The closure laws therefore supply the representation-level congruence needed for the orbit-gap program: once two slices are related by these theorem-forced presentation equivalences, any correct tractability classifier must identify them. The remaining question is whether the resulting closure-hull classifier can be recognized inside the finite structural regime.

Proposition 2.25[25] (The Finite Structural Class Is Nonempty). *Definition 2.19 is nonempty. There exist predicates in the finite structural class; in fact both the constant-true and constant-false slice predicates belong to it.*

Proof sketch. Both predicates are decidable in constant time and are automatically invariant under all closure laws. Their graph extractors are fixed graphs independent of the input slice. For bounded-pattern definability, use a single impossible local pattern: a radius-zero pattern with two distinct vertices cannot occur in any slice. Taking that pattern as forbidden yields the constant-true predicate, and taking it as a required witness yields the constant-false predicate. ■

The finite structural class already contains explicit predicates. The no-go theorem says that exact correctness across the obstruction families is incompatible with closure-orbit agreement inside this class.

Proposition 2.26[26] (Bounded Distinct Action Profiles Compress to Bounded Actions). *For a binary pairwise slice U, let $d(U)$ be the number of distinct action utility profiles. Then there exists a compressed slice U^{prof} with exactly $d(U)$ actions such that a coordinate set is sufficient for U if and only if it is sufficient for U^{prof}, and a coordinate is relevant for U if and only if it is relevant for U^{prof}. Consequently, if $d(U) \le k$, the bounded-actions tractability theorem applies to U^{prof}, so the bounded-distinct-profile subcase is polynomial on the exact-certification side.*

[23] : FR316-328 [24] : FR326-336 [25] : FR200 [26] : FR205-209

Proof sketch. Replace each action by its full utility profile over states and quotient the action set by profile equality. Duplicated actions collapse to a single profile action, but exact certification is unchanged because optimizer equality depends only on which profiles are optimal, not on how many labels realize a given profile. The compressed slice has exactly $d(U)$ actions by construction, and the sufficient-coordinate and relevant-coordinate predicates agree between U and U^{prof}. The existing bounded-actions polynomial-time theorem then applies to the compressed slice. ■

Proposition 2.26 isolates a genuine positive tractable subcase by compression to bounded actions. It is stated as a compression theorem rather than direct membership in the finite structural class because Definition 2.19 fixes a bounded-pattern action alphabet in advance. The proposition shows that nontrivial positive classification work is available through closure-law-respecting reduction inside the framework.

Proposition 2.27[27] (Bounded-Pattern Predicates Stabilize Above a Finite Action Bound). *For every bounded-pattern definable predicate Q, there exists a finite bound B such that Q is constant on all slices with more than B actions. Equivalently, once the action alphabet exceeds the action bound built into the defining pattern scheme, the scheme can no longer distinguish between such slices.*

Proof sketch. In a bounded-pattern scheme every witness and forbidden pattern has action alphabet size at most a fixed bound B. If a slice has more than B actions, then no listed local pattern can occur in it, because occurrence requires an action-set equivalence between the pattern and the slice. The witness branch of the scheme is therefore impossible, while the forbidden branch is determined only by whether the forbidden list is empty. Hence the predicate is constant above the fixed action bound. ■

Proposition 2.26 is stated as a compression theorem rather than as direct admissibility membership for the distinct-profile predicate. Unbounded profile counting is a global algebraic quantity and does not naturally fit a definition whose bounded-pattern clause fixes a finite action alphabet in advance.

Proposition 2.28[28] (Deterministic Payload Transfer). *Let S be a coordinate space, let $\phi : S \to T$ be any deterministic payload map into a finite label type, and let D_ϕ be the induced decision problem with action space T and utility*

$$U(a, s) = \begin{cases} 1, & a = \phi(s), \\ 0, & a \neq \phi(s). \end{cases}$$

Then every coordinate set is sufficient for ϕ if and only if it is sufficient for D_ϕ, and every coordinate is relevant for ϕ if and only if it is relevant for D_ϕ. Equivalently, exact feature sufficiency and relevance for any deterministic payload reduce definitionally to exact relevance certification for an induced decision problem.

Proof sketch. For the induced decision problem, the optimizer at state s is the singleton $\{\phi(s)\}$. Hence two states have the same optimizer if and only if they carry the same payload value. The sufficiency and relevance predicates are therefore literally the same coordinate conditions on S in both formulations. ■

Deterministic payload sufficiency and relevance are exactly exact relevance certification for the induced decision problem.

[27] : `FR214-216` [28] : `FR217-222`

Corollary 2.29[29] (Boolean Payload Transfer). *Let $\phi : S \to \{0,1\}$ be any Boolean payload. Then exact feature sufficiency and relevance for ϕ reduce definitionally to exact relevance certification for the induced decision problem.*

Proof sketch. This is Proposition 2.28 specialized to the two-label codomain $\{0,1\}$. ■

Corollary 2.30[30] (Predicate Transfer). *Let $P(s)$ be any exact yes/no correctness predicate on the state space with decidable truth value. Writing its truth value as a Boolean payload, exact feature sufficiency and relevance for P reduce definitionally to exact relevance certification for the induced decision problem.*

Proof sketch. Encode P by its Boolean truth-value map and apply Corollary 2.29. ■

Corollary 2.31[31] (Nonempty Set-Valued Payload Transfer). *Let $F(s) \subseteq A$ be a nonempty feasible-action set at each state of a coordinate space, and equip allowed actions with utility u_{allowed} and blocked actions with utility u_{blocked} where $u_{\text{blocked}} < u_{\text{allowed}}$. Then exact feature sufficiency and relevance for the set-valued payload F are equivalent to exact relevance certification for the induced decision problem whose optimal set at state s is exactly $F(s)$.*

Proof sketch. The strict utility gap makes the optimizer set coincide with the feasible-action set at every state, so the same coordinate conditions define sufficiency and relevance on both sides. ■

Corollary 2.31 covers total search semantics stated as nonempty feasible-witness sets. Deterministic payloads are the singleton special case.

Corollary 2.32[32] (Arbitrary Set-Valued Payload Transfer via Failure Tokens). *Let $F(s) \subseteq A$ be any set-valued payload, possibly empty. Form the totalized payload on* $\mathrm{Option}(A)$ *by adjoining a distinguished failure token whenever $F(s) = \varnothing$. Then exact feature sufficiency and relevance for F are equivalent to exact relevance certification for the induced decision problem on the totalized payload.*

Proof sketch. Map each admissible output $a \in F(s)$ to `some`(a), and use `none` exactly when $F(s)$ is empty. Equality of the original set-valued payloads is then equivalent to equality of the totalized nonempty payloads, so Corollary 2.31 applies after this totalization step. ■

Corollary 2.32 removes the empty-fiber caveat for search-style semantics. Set-valued payloads with failure states transfer without choosing a deterministic witness selector.

Corollary 2.33[33] (Arbitrary Exact Output Semantics Transfer). *Let $R(s,a)$ be any state-indexed admissible-output relation. Then exact feature sufficiency and relevance for the induced output semantics reduce to exact relevance certification for a decision problem obtained by totalizing the admissible-output sets with a failure token and assigning a strict allowed-versus-blocked utility gap.*

Proof sketch. Write $F(s) = \{a : R(s,a)\}$. This is exactly the arbitrary set-valued case of Corollary 2.32. The relation form is only a notational repackaging of the same theorem. ■

Proposition 2.34[34] (Exactness Means Exact Agreement With Validity). *Let $V(s,a)$ be any state-indexed validity relation. Then the exact relevance profile induced by V is realized by exact relevance certification for the induced decision problem. In particular, exactness refers to exact agreement with V itself, not to zero-error determinism or the absence of approximation, randomization, statistical thresholds, or failure states in the specification.*

[29] : `FR340` [30] : `FR341` [31] : `FR223-225` [32] : `FR226-228` [33] : `FR229-231` [34] : `FR346`

Proof sketch. Corollary 2.33 identifies the sufficient-coordinate family and relevant-coordinate set of the validity semantics with those of the induced decision problem. ∎

Approximation, statistical, and randomized semantics remain exact in the following sense: the specification written into V may be approximate or randomized, but agreement with V is exact.

Rigorous specification, in the exact sense relevant here, is the determination of such a validity relation. If a purported specification does not determine which outputs are valid at each state, objective correctness is not yet fixed, so the exact semantic questions studied here are not well-posed for it. Formal methods do not add this structure from outside; they make explicit the structure that rigor already requires. Different coordinate presentations may make quotient recovery trivial or rich, but they do not change the canonical quotient carried by the specification.

Proposition 2.35[35] (Sufficiency Is Relation Refinement). *Let $R(s, a)$ be any exact correctness relation, write $\mathrm{Adm}_R(s) = \{a : R(s, a)\}$, and define*

$$s \sim_R s' \iff \mathrm{Adm}_R(s) = \mathrm{Adm}_R(s').$$

Then a coordinate set I is sufficient if and only if agreement on I forces $s \sim_R s'$.

Proof sketch. The relation being refined is the admissible-output equivalence relation itself. ∎

Corollary 2.36[36] (Relevance Is Erased Failure of Refinement). *Let $R(s, a)$ be any exact correctness relation. A coordinate i is relevant if and only if agreement on all coordinates except i does not force equality of admissible-output classes. Equivalently, the full coordinate set with i erased is not sufficient.*

Proof sketch. Erase coordinate i and apply Proposition 2.35 to the remaining agreement relation. ∎

Theorem 2.37[37] (Exact Semantics Quotient Universality). *Let $R(s, a)$ be any exact correctness condition, write $\mathrm{Adm}_R(s) = \{a : R(s, a)\}$, and define*

$$s \sim_R s' \iff \mathrm{Adm}_R(s) = \mathrm{Adm}_R(s').$$

Then $\sim_R$ is the canonical semantic object of the problem. Exact relevance certification for the induced decision problem realizes exactly the same sufficient-coordinate family and relevant-coordinate set; a coordinate set is sufficient exactly when it recovers the $\sim_R$-classes; and a coordinate is relevant exactly when erasing it destroys that recovery.

Proof sketch. Corollary 2.33 reduces the exact semantics to exact relevance certification for the induced decision problem. Proposition 2.35 identifies sufficiency with recovery of the admissible-output classes, and Corollary 2.36 identifies relevance with failure of that recovery after erasure. ∎

Corollary 2.38[38] (Universal Scope Over Rigorously Specified Problems). *Every rigorously specified computational problem determines an exact admissible-output semantics, admits a coordinate presentation, and has its exact relevance profile realized by exact relevance certification for the induced decision problem. Sufficiency recovers the admissible-output equivalence classes, and relevance is erased failure of that recovery.*

Proof sketch. Combine Theorem 2.37 with Proposition 2.40. ∎

[35] : FR342 [36] : FR343 [37] : FR345 [38] : FR351

Informally: once correctness is fixed, relevance asks which coordinates recover what matters.

At the semantic level:

$$\text{rigorous specification} \rightsquigarrow \mathrm{Adm}_R \rightsquigarrow \sim_R \rightsquigarrow \text{exact relevance certification.}$$

Proposition 2.39[39] (Canonical Exact Relevance Profile). *Every state-indexed admissible-output relation induces a canonical exact relevance profile, consisting of its sufficient-coordinate family and relevant-coordinate set. This profile is realized by exact relevance certification for the induced decision problem, and it depends only on admissible-output equivalence.*

Proof sketch. Corollary 2.33 identifies the sufficient-coordinate family and relevant-coordinate set of the semantics with those of the induced exact-certification decision problem. The quotient-level theorems then show that these two objects depend only on admissible-output equivalence. ■

Proposition 2.40[40] (Every Exact Specification Admits a Coordinate Presentation). *Every exact output specification admits a coordinate presentation. In the weakest form, one may take the entire state as a single coordinate. Under this presentation, the exact relevance profile of the specification is realized by exact relevance certification for the induced decision problem.*

Proof sketch. Use a one-coordinate presentation whose unique coordinate is the state itself. Agreement on that coordinate is equality of states, so the coordinate presentation is exact. The resulting exact relevance profile agrees with the one carried by the induced exact-certification decision problem. ■

Proposition 2.41[41] (Finite Exact Specifications Admit Coordinate Presentations). *Every finite-state exact output specification admits a coordinate presentation. Concretely, if the state space is finite, one may index coordinates by states and use state-indicator bits as the coordinate map. Under this presentation, the exact relevance profile of the specification is realized by exact relevance certification for the induced decision problem. This strengthens Proposition 2.40 by replacing the trivial one-coordinate presentation with a finite Boolean one.*

Proof sketch. For a finite state space, assign one Boolean coordinate to each state and record whether the current state is that state. Agreement on all coordinates is then equality of states, so the presentation is exact. This indicator-coordinate presentation yields exactly the exact relevance profile carried by the induced decision problem. ■

Proposition 2.42[42] (Countable Exact Specifications Admit Countable Boolean Presentations). *Every exact output specification on an encodable state space admits a countable Boolean presentation. If the state space is countable and carries the discrete measurable or discrete topological structure, this presentation can be chosen measurable or continuous, respectively.*

Proof sketch. Encode each state by a natural number and use its binary digits as Boolean coordinates indexed by $\mathbb{N}$. Equality of all bits is equality of encodings, hence equality of states. In the discrete measurable and discrete topological settings, each coordinate map is measurable or continuous automatically. ■

[39] : FR302-303 [40] : FR307-308 [41] : FR304-306 [42] : FR311-313

Proposition 2.43[43] (Finite Exact Specifications Admit Low-Dimensional Boolean Presentations). *Every finite-state exact output specification admits a Boolean coordinate presentation using only* $\mathrm{size}(|S|)$ *coordinates, where* $\mathrm{size}(|S|)$ *is the bit-length of the state-space cardinality. Under this presentation, the exact relevance profile is again realized by exact relevance certification for the induced decision problem.*

Proof sketch. Index states by $0, \ldots, |S| - 1$ and use the binary digits of the index as Boolean coordinates. Since every index is less than $2^{\mathrm{size}(|S|)}$, equality of those finitely many bits determines the state. This low-dimensional Boolean presentation identifies the induced exact relevance profile with exact relevance certification. ■

Every exact correctness claim admits a coordinate presentation, so the question of which coordinates matter is always a quotient-recovery problem for admissible-output classes. This includes decision, counting, search, approximation guarantees, PAC/regret/risk guarantees, anytime or finite-horizon guarantees, randomized-output guarantees, distributional specifications, and any other specification stated by the outputs that count as correct at each state.

Proposition 2.40 gives the universal one-coordinate realization, while Propositions 2.41–2.43 show that the same semantics can also admit finite or countable Boolean realizations. Vacuous and informative encodings are therefore different presentations of one semantic object, not different semantics. For example, SAT appears through its exact yes/no validity predicate, sorting through the set of correctly sorted outputs for each input, and fixed approximation or PAC guarantees through the outputs meeting the stated threshold.

This universality does not dilute the frontier theorem. It sharpens it. The canonical optimizer-set exact specifications of binary pairwise slices are rigorously specified problems inside the same semantic universe, so any universal exact-certification characterization must already handle that witness class and inherits the same no-go on restriction; see Corollary 6.17 below.

Low-dimensional Boolean presentation and binary pairwise presentation are different notions. The universal reduction above proves the former, not the latter. No universal pairwise normal-form theorem is used anywhere in the frontier argument. The universal no-go works by witness restriction: the binary pairwise obstruction class is already internal to the universal semantic framework, so a treatment claiming universal scope must already survive restriction to that class.

Corollary 2.44[44] (Statistical Guarantee Semantics Transfer). *Let* $R(s, a)$ *specify the admissible outputs for a PAC guarantee, regret guarantee, statistical risk guarantee, anytime guarantee, or finite-horizon guarantee at state* s*. Then exact feature sufficiency and relevance for that guarantee semantics reduce to exact relevance certification for the induced decision problem.*

Proof sketch. Each case is a named specialization of Corollary 2.33. The output object may be a hypothesis, learner, estimator, or policy; correctness still means exact agreement with the admissible-output relation specifying the guarantee, not absence of statistical error thresholds in that guarantee. ■

Corollary 2.45[45] (Randomized-Output Guarantee Semantics Transfer). *Let* $R(s, a)$ *specify the admissible randomized outputs for a state* s*, where* a *may itself be a distribution, kernel, randomized estimator, or randomized policy. Then exact feature sufficiency and relevance for that randomized-output semantics reduce to exact relevance certification for the induced decision problem.*

[43] : FR314-315 [44] : FR268-282 [45] : FR291-293

Proof sketch. Randomization changes the output object, not the meaning of correctness: correctness still means exact agreement with the admissible-output relation on those randomized outputs, not deterministic singleton output semantics. ■

Corollary 2.46 (Randomized-Output Guarantee Semantics Inherit the Closure-Orbit Consequences). *Once a randomized-output guarantee is written as a state-indexed admissible-output relation, the same closure-orbit agreement and no-go theorems apply to classifiers for that semantics. In particular, Theorem 2.20, Corollary 6.14, and Corollary 6.23 apply unchanged after the transfer of Corollary 2.45.*

Proof sketch. The randomized-output instance follows the same transfer-plus-application pattern used above. Corollary 2.45 reduces the semantics to exact relevance certification for the induced decision problem, and the dedicated randomized application theorems then import the same closure-orbit agreement and orbit-gap consequences. ■

Corollary 2.47[46] (Statistical Guarantee Semantics Inherit the Closure-Orbit Consequences). *Once a PAC, regret, statistical-risk, anytime, or finite-horizon guarantee is written as a state-indexed admissible-output relation, the same closure-orbit agreement and no-go theorems apply to classifiers for that semantics. In particular, Theorem 2.20, Corollary 6.14, and Corollary 6.23 apply unchanged after the transfer of Corollary 2.44.*

Proof sketch. Corollary 2.44 reduces each named statistical guarantee semantics to exact relevance certification for the induced decision problem. Any semantics whose tractability predicate is definitionally equivalent to a closure-law-invariant target inherits the same closure-orbit agreement and orbit-gap consequences. Statistical guarantee semantics are named instances of that principle. They also inherit the same exact-semantics quotient invariance, because their sufficient-coordinate family and relevant-coordinate set depend only on admissible-output equivalence. ■

Corollary 2.48[47] (Exact Approximation Semantics Transfer). *Let $R(s, a)$ specify the admissible outputs for an exact approximation specification at state s, for example the set of all outputs meeting a fixed approximation guarantee. Then exact feature sufficiency and relevance for that approximation semantics reduce to exact relevance certification for the induced decision problem.*

Proof sketch. Approximation enters only through the specification: once the admissible-output relation says which outputs meet the guarantee, correctness with respect to that approximation specification is exact agreement with that relation, not zero approximation error. ■

Table 2 summarizes the named exact-semantics reductions established in this section.

Proposition 2.49[48] (Approximate Relevance and Sufficiency Claims Need Explicit Stability Control). *Let D and D' be decision problems on the same coordinate space, and suppose their utilities are uniformly δ-close. If coordinate i has a relevance witness in D given by states s, s' with distinct strict optimal actions at the two witness states, and if the strict utility gap at each witness state exceeds 2δ, then i is also relevant in D'. Likewise, if a coordinate set I has a non-sufficiency witness in D given by states s, s' that agree on I but have distinct strict optimal actions, and if the strict utility gap at each witness state exceeds 2δ, then I is still not sufficient in D'.*

Proof sketch. The existing uniform-approximation theorem preserves the optimizer set at any state whose strict utility gap dominates the approximation error. Applying that theorem to both witness states keeps the two singleton optimizer sets distinct. In the relevance case this preserves the same

[46] : `FR291-296` [47] : `FR232-234` [48] : `FR263`, `FR266`

Specification Type	Exact Semantics	Exact-Certification Equivalent
Deterministic payload $\phi(s)$	singleton payload $\{\phi(s)\}$	induced decision problem with $\mathrm{Opt}(s) = \{\phi(s)\}$
Boolean predicate $P(s)$	truth-value payload in $\{0, 1\}$	Boolean-payload instance of the induced decision problem
Set-valued search $F(s)$	feasible-output set $F(s) \subseteq A$	induced decision problem with optimizer set exactly $F(s)$
Relational output $R(s, a)$	admissible-output set $\{a : R(s, a)\}$	totalized decision problem with failure token and strict allowed/blocked gap
Approximation or PAC/regret/risk guarantee	outputs satisfying the stated guarantee	named instance of the same relational-output reduction
Randomized output	admissible distributions, kernels, estimators, or policies	randomized-output instance of the same relational-output reduction

Table 2: Exact semantic reductions to quotient recovery. The named transfer corollaries differ only in how the admissible-output relation is packaged.

relevance witness. In the sufficiency case the same state pair still agrees on I but still has different optimizer sets, so it remains a non-sufficiency witness. ■

Corollary 2.50[49] (Arbitrarily Small Uniform Perturbations Can Flip Relevance). *For every $\varepsilon > 0$, there exist two decision problems on the same one-coordinate Boolean state space that are uniformly ε-close, yet the unique coordinate is relevant in one problem and irrelevant in the other.*

Proof sketch. An explicit two-action, one-coordinate construction gives the result. In the first problem the optimizer tracks the Boolean state, so the unique coordinate is relevant. In the second problem all actions are tied at every state, so the coordinate is irrelevant. The two utility functions differ uniformly by at most ε. ■

Corollary 2.51[50] (Arbitrarily Small Uniform Perturbations Can Flip Sufficiency). *For every $\varepsilon > 0$, there exist two decision problems on the same one-coordinate Boolean state space that are uniformly ε-close, yet the empty coordinate set is sufficient in one problem and not sufficient in the other.*

Proof sketch. The same one-coordinate construction separates the constant-optimizer case from the state-tracking case. In the flat problem the empty set is sufficient because the optimizer set is constant, while in the state-tracking problem the empty set is not sufficient because the two Boolean states induce different optimizer sets. The two utility functions are still uniformly ε-close. ■

Proposition 2.52[51] (Global Approximation Stability Under Uniform Strict Gaps). *Let D and D' be decision problems on the same coordinate space, and suppose their utilities are uniformly δ-close. If every state of D has a strict optimal action whose utility gap exceeds 2δ, then D and D' have the same optimizer quotient, the same sufficient-coordinate family, the same relevant-coordinate set, and the same minimal sufficient sets.*

[49] : `FR264` [50] : `FR267` [51] : `FR288-301`

Proof sketch. The uniform-approximation theorem preserves the optimizer set at each state once the local strict utility gap dominates the approximation error. Applying this statewise yields equality of optimizer sets for all states. The optimizer quotient, sufficient-coordinate family, relevant-coordinate set, and minimal sufficient sets therefore coincide. ■

Thus closeness alone does not justify an approximate relevance or sufficiency claim. Any ε-based surrogate claim needs an explicit reduction to the exact optimizer sets, for example by a witness-gap bound as in Proposition 2.49, because otherwise arbitrarily small perturbations can change the exact judgment.

Informally: without a margin, approximation does not preserve what matters.

Proposition 2.53[52] (Exact Semantic Claims Depend Only on Admissible-Output Equivalence). *Let R and R' be two exact output semantics on the same state space. If they induce the same equality relation on admissible-output sets, then they induce the same sufficient coordinate sets and the same relevant coordinates. Exact semantic claims about sufficiency and relevance therefore depend only on the induced admissible-output equivalence relation.*

Proof sketch. The transfer theorems reduce both semantics to exact relevance certification for induced decision problems. If the admissible-output equality relation is the same, the induced sufficiency and relevance structures are the same as well. ■

Corollary 2.54[53] (Every State Equivalence Relation Is Realizable as Exact Output Semantics). *For every equivalence relation on the state space, there exists an exact output semantics whose admissible-output equality relation is exactly that equivalence relation.*

Proof sketch. Take the output space to be the quotient by the given equivalence relation and assign to each state its own quotient class as a singleton admissible-output set. Equality of those singleton output sets is then exactly the original equivalence relation. ■

Proposition 2.55[54] (Semantically Extensional Claims Factor Through the Exact-Semantics Quotient). *Let C be any claim about exact output semantics that depends only on admissible-output equivalence. Then C factors through the quotient of output semantics by admissible-output equivalence. In particular, once a semantic claim is exact and extensional, the exact-semantics quotient is the object of the claim.*

Proof sketch. Semantically extensional exact claims descend to the quotient by admissible-output equivalence. If C takes the same truth value on any two semantics with the same admissible-output equivalence, then C factors through that quotient. ■

Closure-law invariance is theorem-forced. Theorem 2.20 shows that correctness already forces closure-orbit agreement, and Theorem 6.11 together with Proposition 6.12 shows that this theorem-forced congruence is the only hypothesis used in the no-go theorem. Clauses 1, 3, and 4 are guardrails against the failure modes identified in Proposition 2.17.

[52] : `FR235-236` [53] : `FR237`, `FR240` [54] : `FR241-244`

3 Finite Basis for the Current Positive Landscape

We distinguish between *tractable families treated here* and *primitive tractability mechanisms*. A tractable family is any named positive condition included in this inventory. A primitive mechanism is a smaller conceptual source of tractability that may explain several families at once. The positive theory organizes into three roles: six core structural families, four regime lifts, and five degenerate collapses. Projecting those families to primitive mechanisms leaves eight sources of tractability: the six core subcases together with constant-optimizer collapse and finite explicit enumeration. These eight mechanisms are the explicit inventory of the current positive landscape, not a complete classification of all tractability mechanisms for exact relevance certification. The universal theorem is quotient universality for exact correctness; the later no-go shows that no finite structural classifier extends this positive inventory to an exact frontier theorem.

Theorem 3.1[55] (Explicit Partition of the Current Positive Landscape). *The fifteen tractable families considered here admit an explicit disjoint partition into three blocks:*

1. *six* core *structural families,*
2. *four* lifted *families that reduce to existing core subcases, and*
3. *five* degenerate *families that become tractable by optimizer collapse or finite explicit enumeration.*

Moreover, each family is classified into exactly one of these three blocks by an explicit classification map.

Proposition 3.2[56] (Explicit Inventory). *The fifteen tractable families considered here are exactly the following:*

1. *core: bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry;*
2. *lifts: product distribution, bounded support, bounded horizon, full observability;*
3. *degenerate: single action, bounded state space, strict global dominance, constant optimal set, multiplicative-separable constant-sign.*

Corollary 3.3[57] (Finite Basis for the Current Positive Landscape). *Projecting the partition of Theorem 3.1 to primitive mechanisms yields an explicit finite list of eight primitive tractability mechanisms. The list consists of the six core structural subcases*

$$\textit{bounded actions},\ \textit{separable utility},\ \textit{low tensor rank},$$
$$\textit{tree structure},\ \textit{bounded treewidth},\ \textit{coordinate symmetry}$$

together with two degenerate mechanisms:

$$\textit{constant-optimizer collapse} \qquad \textit{and} \qquad \textit{finite explicit enumeration.}$$

Proposition 3.4[58] (Explicit Primitive Basis). *The current primitive basis consists exactly of bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry, constant-optimizer collapse, and finite explicit enumeration.*

The current positive landscape therefore contains fifteen tractable families accounted for by eight primitive mechanisms.

Table 3 summarizes the role partition and the induced primitive basis in one place.

[55] : `FR1-2` [56] : `FR25-28` [57] : `FR3` [58] : `FR29`

Role	Family	Primitive Mechanism
Core structural	bounded actions	bounded actions
Core structural	separable utility	separable utility
Core structural	low tensor rank	low tensor rank
Core structural	tree structure	tree structure
Core structural	bounded treewidth	bounded treewidth
Core structural	coordinate symmetry	coordinate symmetry
Regime lift	product distribution	separable utility
Regime lift	bounded support	bounded actions
Regime lift	bounded horizon	bounded treewidth
Regime lift	full observability	tree structure
Degenerate	single action	constant-optimizer collapse
Degenerate	strict global dominance	constant-optimizer collapse
Degenerate	constant optimal set	constant-optimizer collapse
Degenerate	multiplicative-separable constant-sign	constant-optimizer collapse
Degenerate	bounded state space	finite explicit enumeration

Table 3: Partition of the tractable families treated here. The primitive-mechanism column is the projection used in Corollary 3.3.

Proposition 3.5[59] (All Independent Core Mechanisms Are Nondegenerate). *The analysis already contains witness families showing that five core mechanisms are individually indispensable in the sense of basis coverage:*

1. *a low-rank-only witness family,*
2. *a separable-only witness family,*
3. *a bounded-actions-only witness family,*
4. *a bounded-treewidth-only witness family, and*
5. *a symmetry-only witness family.*

Each of these families remains tractable by its designated mechanism while failing the corresponding versions of the other independent core mechanisms.

Proposition 3.5 settles the independent part of the basis question on the present surface. Every core mechanism that is not merely contained in another is shown to be genuinely needed.

Concretely, the present theory contains five kinds of "only-by" witness: low-rank only, separable only, bounded-actions only, bounded-treewidth only, and symmetry only.

Proposition 3.6[60] (Weak Tree Ordering Does Not Imply Width One). *There exists a dependency presentation satisfying the older monotone tree-order predicate* `TreeStructured` *whose dependency*

[59] : `FR101`, `FR105-106`, `FR108`, `FR112-113` [60] : `FR114`

graph has real treewidth greater than one. Width-one containment requires the stricter parent-tree condition from Proposition 3.8, not merely monotone ordering.

Proposition 3.7[61] (Cyclic Dependencies Recover Hardness). *The reduction family for exact relevance certification yields* coNP*-hard instances with cyclic dependency structure. Thus the tree-structured regime is not only sufficient for tractability; leaving it also reaches the hard side.*

Proposition 3.8[62] (Parent-Tree Structure Gives Width-One Decompositions). *Suppose a dependency presentation is tree-structured in the strict parent sense: every dependency points to a smaller coordinate and every coordinate has at most one parent. Then the induced dependency graph admits a genuine tree decomposition of width at most one.*

Proposition 3.8 is the containment theorem behind the intended "tree structure is the treewidth-one special case" slogan. It distinguishes this parent-tree notion from the weaker monotone-order predicate currently named `TreeStructured`.

Proposition 3.9[63] (Current Positive Interfaces Factor Through Role Classification). *The current positive theorem surface factors through the role classifier of Theorem 3.1. Concretely, the family-indexed interfaces for*

1. *subcase-facing tractability,*
2. *degenerate-case witnesses, and*
3. *complexity-class summaries*

all factor through the map

$$\textit{family} \longmapsto \textit{role} \in \{\textit{core}, \textit{lifted}, \textit{degenerate}\}.$$

Proposition 3.9 shows that the exported positive API is already controlled by the role partition rather than by fifteen unrelated theorem schemas.

4 Lifts and Degeneracies

The extra named families beyond the six core subcases split naturally into two groups.

Regime Lifts

Four stochastic/sequential positive results do not introduce new primitive mechanisms. They reduce directly to existing core subcases:

1. product distributions reduce to separable utility;
2. bounded support reduces to bounded actions;
3. bounded horizons reduce to bounded treewidth;
4. full observability reduces to tree structure.

These are mathematically useful because they transfer the static tractable theory into richer regimes. But they do not enlarge the primitive basis. They are best understood as *lifts*, not new frontier points.

Proposition 4.1[64] (Lifted Cases Introduce No New Primitive). *Each stochastic/sequential tractable case considered here is classified into one of the six core structural subcases. In particular, these positive results do not enlarge the primitive basis from Corollary 3.3.*

[61] : `FR115` [62] : `FR100, FR110-111` [63] : `FR4` [64] : `FR5`

Degenerate Cases

The remaining positive cases considered here are degenerate in a precise sense.

Single-action systems, strict global dominance, constant optimal-set systems, and multiplicative-separable constant-sign models all collapse the optimizer so strongly that the relevant-coordinate question disappears: the optimal-action set is constant across states, so the empty coordinate set is sufficient. These are not new structural tractability mechanisms for the relevance problem; they are constant-optimizer collapses.

Bounded state space behaves differently. It does not force a constant optimizer. Instead, it makes exact certification feasible by brute-force enumeration over a finite universe. This is polynomial only because the instance family already carries a hard external bound on its search space.

Proposition 4.2[65] (Degenerate Positive Cases). *Every non-core tractable case considered here that is not a regime lift belongs to one of two degenerate buckets:*

1. *constant-optimizer collapse, or*
2. *finite explicit enumeration.*

Taken together with Proposition 4.1, Proposition 4.2 sharpens Theorem 3.1: every positive result considered here is exactly one of three things, a primitive core case, a lift, or a degenerate collapse.

5 Realizability of Arbitrary Quotients

One possible route to a finite tractability taxonomy would be to show that optimizer-induced quotients realize only a narrow subclass of equivalence relations, so that the frontier would emerge from realizability alone. In the unconstrained setting, this route fails for a simple reason: realizability is already maximal.

Theorem 5.1[66] (Every Labeling Kernel Is Optimizer-Realizable). *Let $\phi : S \to T$ be any function. There exists a decision problem with action space T and state space S such that for every state $s \in S$,*

$$Opt(s) = \{\phi(s)\}.$$

Consequently, for all states $s, s' \in S$,

$$Opt(s) = Opt(s') \iff \phi(s) = \phi(s').$$

Equivalently, the decision quotient of the constructed problem is exactly the kernel partition of ϕ.

Proof. Take the action set to be T itself and define

$$U(a, s) = \begin{cases} 1 & \text{if } a = \phi(s), \\ 0 & \text{otherwise.} \end{cases}$$

Then the unique maximizer at state s is the designated action $\phi(s)$, so the optimizer set is exactly $\{\phi(s)\}$. Two states therefore have the same optimizer set if and only if they receive the same label under ϕ. ■

[65] : FR1, FR6 [66] : FR7-8

Theorem 5.1 eliminates quotient shape as the organizing principle for a frontier theorem. The construction is elementary, and optimizer realizability by itself is extremely permissive. It also gives a representational universality statement: any classification problem presented as a labeling kernel already embeds into optimizer-quotient form. If arbitrary label kernels are already optimizer-induced, then no finite taxonomy can come merely from asking which abstract quotients are realizable.

Informally: quotient shape alone does not determine what matters for tractability.

Proposition 5.2[67] (Every Equivalence Relation Is Optimizer-Realizable). *For every equivalence relation on the state space, there exists a decision problem whose decision quotient relation is exactly that equivalence relation. Moreover, the quotient object of the realizing problem is canonically equivalent to the corresponding setoid quotient.*

Proof. Let $\approx$ be an equivalence relation on S, and let $\pi : S \to S/{\approx}$ be the quotient map. Apply Theorem 5.1 to the labeling map π. The resulting decision problem satisfies

$$\mathrm{Opt}(s) = \mathrm{Opt}(s') \iff \pi(s) = \pi(s'),$$

and the right-hand side is exactly the original equivalence relation. Since the realizing quotient identifies states precisely by equality of quotient labels, its quotient object is canonically equivalent to the setoid quotient itself. ∎

Proposition 5.2 is the maximal realizability statement available in the framework. The labeling-kernel theorem is a special case. At the level of abstract quotient relations, realizability is maximal.

Proposition 5.3[68] (The Realized Quotient Is Exactly the Label Range). *For every labeling map $\phi : S \to T$, the decision quotient of the realizing problem is canonically equivalent to the range of ϕ. In finite settings, the quotient cardinality is therefore exactly $|\operatorname{range}(\phi)|$.*

Proof. By Theorem 5.1, two states lie in the same decision class if and only if they receive the same label under ϕ. Therefore the quotient class of a state depends only on the value $\phi(s)$, and sending the class of s to $\phi(s)$ defines a well-defined map from the decision quotient to $\operatorname{range}(\phi)$. This map is surjective by definition of the range and injective because equality in the range means equality of labels, hence equality of quotient classes. ∎

Proposition 5.3 strengthens Theorem 5.1. The realizing construction does not merely realize the induced equivalence relation abstractly; it realizes the quotient object itself with exactly the expected image size.

Corollary 5.4[69] (Realizability Alone Cannot Be the Frontier). *Any future optimizer-compatible tractability metatheorem must use more than quotient realizability alone. Additional restrictions have to enter through representation, coordinate structure, utility structure, closure properties of the family under study, or comparable algebraic invariants.*

This shifts the burden of the metatheorem. Quotient shape does not constrain the frontier. The remaining question is which *natural structural families of decision problems* force tractability or preserve hardness once coordinate structure and utility representation are taken seriously, and which natural restrictions on decision problems constrain the quotient geometries that can occur.

[67] : `FR45-47` [68] : `FR39-40` [69] : `FR7-8`

6 Toward the Frontier Theorem

Sections 3–5 motivate a frontier question. Do the tractable families already suggest a finite optimizer-compatible criterion? In the strongest form, could exact relevance certification admit a representation-sensitive dichotomy theorem analogous to the classical dichotomy programs for satisfiability and constraint satisfaction?

The most direct version of that program fails. Quotient shape is too expressive, and later in the section even structurally restricted finite classifiers fail on a theorem-forced surface. The negative theorem proved here is correspondingly scoped: it rules out exact classifiers in the direct closure-invariant structural regime, not every possible representation-sensitive invariant.

Three obstacles remain. Realizability is not the bottleneck: Theorem 5.1 shows that arbitrary labeling kernels already arise as optimizer quotients, so no frontier theorem can be driven by quotient realizability alone. Even on the positive side one must separate primitive mechanisms from lifts and degeneracies; otherwise a finite inventory confuses genuine sources of tractability with derived or collapsed cases. And although the long-term target remains an algebraic description analogous to the CSP dichotomy program, the negative results below show that no direct optimizer-compatible classifier can play that role.

A Stabilized Binary Pairwise Subregime

Binary pairwise slices are the smallest representation class in which unary collapse, dense interaction, optimizer degeneracy, and the orbit-gap obstruction all already appear in full.

One piece of the frontier program already admits a clean canonical theorem. On binary coordinate domains, define the mixed difference along coordinates i, j for action a by evaluating the utility on the four assignments $(0,0),(1,1),(0,1),(1,0)$ while holding all other coordinates fixed at 0. For pairwise utilities, this mixed difference is the canonical witness of genuine pair interaction.

Proposition 6.1[70] (Binary Pairwise Symmetry Dichotomy). *Let the coordinate alphabet be binary. If a utility admits a pairwise decomposition and is invariant under coordinate permutations, then either:*

1. *it admits a unary-coordinate decomposition, so every pairwise term collapses into single-coordinate contributions, or*
2. *every distinct coordinate pair carries a genuine pair interaction, witnessed by a nonzero mixed difference for some action.*

Within the binary pairwise-symmetric regime, there is no intermediate sparse interaction pattern hiding between unary collapse and the complete-pair case. Once a single nonzero pair witness exists, symmetry propagates it to every coordinate pair. This removes one plausible source of additional primitive mechanisms from the frontier search.

The optimizer-relevant version needs one extra layer. Raw pair interaction is still too coarse, because action-independent pairwise terms can be structurally dense without changing the optimizer. The correct invariant is the mixed difference of action gaps $U(a,\cdot) - U(b,\cdot)$.

Proposition 6.2[71] (Decision-Relevant Binary Pairwise Dichotomy). *In the same binary pairwise-symmetric regime, either all action-dependent pair effects collapse into a unary-coordinate reduction after removing an action-independent base state term, or every distinct coordinate pair is decision-relevantly interacting for some action gap.*

[70] : `FR118-120` [71] : `FR121-124`

Proposition 6.3[72] (Symmetric Binary Pairwise Decision-Relevant Graphs Are Either Empty or Complete). *In the same binary pairwise-symmetric regime, the decision-relevant interaction graph is either edgeless or complete.*

Within this regime the decision-relevant graph therefore collapses to a two-point algebra. Later in the subsection, action-dependent, state-independent utility offsets are shown to preserve this graph, so the offset-normalized decision-relevant graph is a concrete successor-class candidate with existing algebraic control.

A coarse hardness heuristic therefore fails. Even complete genuine pair interaction and failure of coordinate symmetry do not force hardness by themselves: one can still have a constant optimizer. Any eventual frontier theorem must therefore quotient out optimizer-degenerate phenomena, not just declared or utility-level interaction density.[73]

Action-specific utility offsets leave the decision-relevant interaction graph unchanged, yet they can turn a family with nonconstant optimizer behavior into one with a constant optimizer while preserving dense decision-relevant structure. Any true hardness theorem in this regime must therefore normalize away action offsets explicitly, not merely require dense decision-relevant interaction.[74]

Passing to action-dependent, state-independent offset classes makes the offset-normalized decision-relevant graph well-defined, but the dichotomy attempt still fails. One can retain complete decision-relevant interaction and unbounded treewidth while exact certification remains trivial.[75]

Restricting the interaction graph to optimizer-supported actions removes two earlier collapse mechanisms: the offset-collapse family and a ghost-action family whose all-action decision-relevant graph is complete even though a single coordinate remains sufficient. But even that support-filtered graph is not enough. The optimizer-supported obstruction theorem exhibits a margin-masking family with complete optimizer-supported decision-relevant interaction and unbounded treewidth, yet the optimizer still depends only on coordinate 0. The third collapse mechanism is therefore not ghost support or offset collapse, but large unary margins that mask dense supported pairwise interactions.[76]

One might hope that a strict unary-to-pair margin bound rescues the dichotomy. Define *margin-bounded* pairwise utilities by requiring every unary term to be at most twice the largest binary mixed-difference magnitude. This still fails. The dominant-pair family is margin-bounded because its unary terms vanish, while its optimizer-supported decision-relevant graph remains complete. Exact certification is still controlled by the two-coordinate set $\{0, 1\}$. The fourth collapse mechanism is therefore pair-weight concentration rather than unary masking: one supported pair dominates the optimizer while the remaining dense interactions are too weak to matter.[77]

Admissible Closure-Orbit No-Go Program

The obstruction families above lead to a uniform negative statement because they admit explicit disagreements inside a single closure orbit.

Theorems 6.6–6.9 concern four representation-level predicates. Corollary 6.14 and Corollary 6.16 concern finite structural access to tractability on the same domain.

The generic statement comes before the family-specific one. Proposition 6.18 identifies orbit gaps as the complete obstruction criterion for exact classification by closure-law-invariant predicates. Theorem 6.11 applies that criterion to any closure-sound predicate class. Corollary 6.14 removes even explicit closure-soundness by deriving closure-orbit agreement from correctness itself. Definition 2.19 specifies the finite structural class used inside that generic framework.

[72] : FR359 [73] : FR124 [74] : FR125-127 [75] : FR128-129 [76] : FR130-136 [77] : FR137-140

Proposition 6.4[78] (Orbit-Gap Template). *Let Q be a slice predicate. Suppose there exist slices U, V such that U and V lie in the same closure orbit, $Q(U)$ holds, and $Q(V)$ fails. Then no closure-law-invariant predicate P can satisfy*

$$P(W) \iff Q(W) \qquad \textit{for all slices } W.$$

Proof idea. Closure-law invariance gives $P(U) \iff P(V)$. Exact agreement with Q would then imply $Q(U) \iff Q(V)$, contradicting the assumed orbit gap. ■

For each obstruction family, the argument constructs two same-orbit slices with different obstruction status. In all four cases the witness is a positive-affine step whose α-component is an action-independent state term supported on a single coordinate pair. Pure relabeling cannot change the relevant obstruction statistics, and a global scale factor preserves all pairwise magnitudes up to common rescaling. The statewise additive term moves mass onto a chosen pair while remaining inside the theorem-forced closure laws. One theorem-forced transport therefore destabilizes several natural candidate predicates. Proposition 6.18 together with Corollary 6.19 shows that orbit gaps are the complete obstruction criterion in the closure-invariant and correctness-forced regimes.

Illustrative Orbit Example

The dominant-pair witness below is the smallest concrete orbit-gap example in the paper. One affine closure step changes the frontier predicate while leaving the underlying exact-certification problem unchanged.

Take three binary coordinates $x = (x_0, x_1, x_2) \in \{0,1\}^3$ and two actions a, b. Define

$$U(a,x) = 2x_0x_1, \qquad U(b,x) = 0.$$

The only nonzero pair interaction occurs on $\{0,1\}$, so U has unique dominant pair $\{0,1\}$. Its optimizer behavior depends only on whether $x_0x_1 = 1$, since

$$U(a,x) - U(b,x) = 2x_0x_1.$$

Now apply one allowed positive-affine closure step with scale 1 and action-independent state term

$$\alpha(x) = 3x_1x_2, \qquad V(c,x) = U(c,x) + \alpha(x) \quad (c \in \{a,b\}).$$

The optimizer sets are unchanged because

$$V(a,x) - V(b,x) = U(a,x) - U(b,x) = 2x_0x_1,$$

so U and V determine the same exact-certification problem and lie in the same closure orbit. But the raw pair statistics have changed: V now has larger pair magnitude on $\{1,2\}$ than on $\{0,1\}$. Thus the dominant-pair predicate flips from $\{0,1\}$ to $\{1,2\}$ even though the optimizer quotient, sufficient sets, and relevant coordinates are unchanged. This is the orbit-gap mechanism in its smallest concrete form.

Remark 6.5 (Transport Interpretation of the Affine Witnesses). Once transport is computed on optimizer-quotient classes, a singleton quotient admits zero-cost diagonal transport, whereas genuine branching forces positive off-diagonal transport when distinct classes carry mass. The pair-targeted affine witnesses exploit exactly this sensitivity: they move mass across a selected branch of the quotient while staying inside the same closure orbit. The witnesses therefore follow the quotient transport geometry rather than functioning as arbitrary syntactic perturbations. In the named families used below, these witnesses are bounded perturbations on constant-size support, so the representation size remains polynomially equivalent under standard sparse encodings.[79]

[78] : FR185 [79] : FR183-184

Theorem 6.6[80] (Dominant-Pair Finite-Structural No-Go). *No predicate in the finite structural class can decide the dominant-pair obstruction family exactly.*

Proof sketch. The target predicate is unique dominant-pair status: one pair/action realizes the maximal interaction magnitude and that maximizer is unique. Start from a slice whose unique dominant pair is $\{0, 1\}$. Add an action-independent affine state term supported on a different pair, say $\{1, 2\}$. This produces a slice in the same closure orbit but with a different unique dominant pair. Proposition 6.4 then rules out every closure-law-invariant classifier, hence every admissible one. ■

Theorem 6.6 gives the first instance. The remaining three families use the same orbit-gap scheme, each with a different target predicate and therefore a different failure mode inside the same closure orbit.

At the compute-cost layer, the same dominant-pair orbit witness yields a machine-checked no-go for optimizer computation: no correct classifier for polynomial-time solvability of optimizer computation can decide dominant-pair status exactly. In particular, no admissible predicate can simultaneously track optimizer-computation polynomiality and exact dominant-pair status.[81]

Theorem 6.7[82] (Margin-Masking Finite-Structural No-Go). *No predicate in the finite structural class can decide margin-boundedness exactly.*

Proof sketch. For margin masking, the target predicate is margin-boundedness itself. This family is structurally different from the dominant-pair family because it is governed by a threshold comparing unary mass against the largest pair interaction. The affine witness leaves unary terms unchanged but raises the largest pair interaction past the relevant threshold, so the translated slice becomes margin-bounded while the base slice is not. Proposition 6.4 therefore rules out every closure-law-invariant classifier, hence every admissible one. ■

Theorem 6.8[83] (Ghost-Action Finite-Structural No-Go). *No predicate in the finite structural class can decide the ghost-action concentration signature exactly.*

Proof sketch. For ghost actions, the target predicate says that there is an action whose unary contribution on the first coordinate is -1 on both binary values and whose anchor-pair interaction has unit mixed-difference magnitude. This family is local in appearance but unstable under the same pair-targeted affine move. A pair-supported affine term preserves the closure orbit while destroying the signature, so Proposition 6.4 applies. ■

Theorem 6.9[84] (Offset Finite-Structural No-Go). *No predicate in the finite structural class can decide the additive/statewise offset signature exactly.*

Proof sketch. For additive/statewise offset concentration, the target predicate requires two actions with distinct anchor-pair interaction magnitudes, namely 1 and 0. This family isolates the action-specific mismatch that survives after earlier offset normalizations. A pair-supported affine term changes the anchor-pair statistics while remaining in the same closure orbit. Again, Proposition 6.4 yields the contradiction. ■

Theorem 6.10[85] (Finite-Structural Collapse Across the Four Obstruction Families). *Let P be a normalization predicate in the finite structural class. Then P is not a tractability characterization.*

[80] : `FR186` [81] : `FR250-251` [82] : `FR187` [83] : `FR188` [84] : `FR189` [85] : `FR190`

Proof idea. If P belonged to the finite structural class and were an exact tractability characterization, it would have to decide each of the four obstruction predicates above. Theorem 6.6, Theorem 6.7, Theorem 6.8, and Theorem 6.9 show that this is impossible. Hence no predicate in the finite structural class yields the desired frontier theorem. ■

The four obstruction families are witnesses for the orbit-gap template (Proposition 6.4), not the scope of Theorem 6.10. Nor are they four different proof mechanisms: the witness move is uniform, while the target predicates differ. The theorem ranges over the entire finite structural class. The role of the four families is to show that the same closure-orbit mechanism destabilizes several natural representation-level predicates, while Proposition 6.18 identifies orbit gaps as the complete obstruction criterion for exact classification by closure-law-invariant predicates.

Table 4 summarizes the target predicates and the corresponding collapse modes.

Obstruction Family	**Target Predicate**	**Collapse Mechanism**
Dominant-pair	unique dominant-pair status	pair-weight concentration with vanishing unary terms
Margin-masking	margin-boundedness	large unary margins mask dense supported pairwise interaction
Ghost-action	ghost-action concentration signature	ghost support masks dense interaction
Additive/statewise offset	additive/statewise offset signature	action-specific offsets collapse interaction after offset-insensitive normalization

Table 4: The orbit-gap witness move is uniform across the four obstruction families; only the target predicate changes.

Abstractly, let Γ be a class of slice predicates, and call a predicate Γ-admissible when it belongs to Γ. Call Γ *closure-sound* if every Γ-admissible predicate is closure-law invariant. Call Γ a *finite structural admissibility package* if it is closure-sound and also imposes auxiliary restrictions such as efficient checkability, structural extractability, and bounded-pattern definability. The proof below uses only closure-soundness; the remaining clauses specify which finite structural search spaces the no-go is meant to cover.[86]

Theorem 6.11[87] (Closure-Sound Package No-Go). *Let Γ be any class of slice predicates such that every Γ-admissible predicate is closure-law invariant. Then no Γ-admissible predicate can simultaneously characterize the four obstruction families. The no-go theorem depends only on closure-law invariance; the remaining clauses of Definition 2.19 delimit the finite structural class and do not enter the contradiction.*

Proof sketch. Closure-law invariance suffices to invoke Proposition 6.4 against each of the four obstruction families. The same orbit-gap proofs underlying Theorem 6.6, Theorem 6.7, Theorem 6.8, and Theorem 6.9 then yield the contradiction directly. Any one of the four families independently witnesses the collapse; we state all four to make explicit that the obstruction is robust across multiple natural target predicates. ■

At the Γ-generic level, the contradiction is simultaneous: one Γ-admissible predicate cannot characterize all four families at once. The individual-family contradictions above are stated separately for the finite structural class of Definition 2.19.

[86] : `FR191-192` [87] : `FR193`

Proposition 6.12[88] (Logical Dependency of the No-Go). *The contradiction in Theorem 6.11 depends only on closure-law invariance. The derivation of closure-law invariance in Theorem 2.20 depends only on correctness over a closure-closed domain together with Proposition 2.18. Polynomial-time checkability, structural extractability, and bounded-pattern definability do not enter either proof.*

Proof. Immediate from the displayed proofs of Theorem 6.11 and Theorem 2.20. ∎

Theorem 6.11 is the generic form of Theorem 6.10. Any admissibility notion whose predicates are closure-law invariant inherits the impossibility conclusion. Proposition 2.1 identifies closure-law invariance as the semantic core, and Theorem 2.20 shows that correctness forces the same invariance on any tractability classifier. Definition 2.19 specifies the finite structural class used in the named family theorems.

Corollary 6.13[89] (Finite Structural Admissibility Packages). *The same conclusion holds for every finite structural admissibility package. In particular, efficient checkability, structural extractability, bounded-pattern definability, and the exclusion of direct semantic encoding restrict the search space but do not enter the contradiction itself.*

Proof. Immediate from Theorem 6.11, since only the closure-soundness clause is used. ∎

Corollary 6.14[90] (No Finite Structural Tractability Classifier). *Let C be a tractability classifier on a domain that is closed under the closure laws and contains the four obstruction families. If the predicate computed by C belongs to the finite structural class, then C cannot yield an exact tractability characterization across those families.*

Proof. If C were correct, Theorem 2.20 would force closure-orbit agreement on its domain. If its predicate also belonged to the finite structural class, Theorem 6.11 would rule out exact characterization across the four obstruction families. ∎

Correctness alone still forces closure-orbit agreement on any closure-closed domain containing the obstruction families. Theorem 6.11 isolates closure-law invariance as the only hypothesis used in the contradiction, Proposition 6.12 identifies the exact dependency of the contradiction, and Theorem 2.20 derives the same orbit agreement directly from correctness. The abstract closure-hull classifier therefore exists whenever tractability is orbit-constant on the domain, but Corollary 6.14 says that this classifier is not available inside the finite structural regime. Corollary 6.23 gives the domain-relative form: restricting the domain helps only if it removes all orbit gaps for the target notion.

Proposition 6.15[91] (Full Binary Pairwise Domain Is Closure-Closed). *Let $\mathcal{U}$ be the class of all binary pairwise slices. Then $\mathcal{U}$ is closed under action relabeling, coordinate relabeling, positive affine reparameterization, action duplication, state duplication, and binary irrelevant-coordinate extension. The dominant-pair, margin-masking, ghost-action, and additive/statewise offset obstruction families are contained in $\mathcal{U}$. Therefore $\mathcal{U}$ is a closure-closed domain containing the four obstruction families.*

Proof sketch. Each theorem-forced presentation move maps binary pairwise slices to binary pairwise slices. The four obstruction families are binary pairwise slices by construction. ∎

The binary pairwise domain is already sufficient for the obstruction. Any larger closure-closed representation class containing $\mathcal{U}$ inherits the same impossibility conclusion by restriction.

[88] : `FR347` [89] : `FR194` [90] : `FR193`, `FR197-199` [91] : `FR348-349`

Corollary 6.16[92] (No Finite Structural Tractability Classifier on the Full Binary Pairwise Domain). *No tractability classifier in the finite structural class yields an exact characterization across the four obstruction families on the full binary pairwise slice domain* $\mathcal{U}$.

Proof. Apply Corollary 6.14 together with Proposition 6.15. ■

On this domain, tractability is constant on closure orbits, so Corollary 6.19 yields an abstract exact closure-invariant classifier for tractability on $\mathcal{U}$. Corollary 6.16 says that no such classifier belongs to the finite structural class. The four obstruction families witness this structural inaccessibility by showing that natural representation-level proxies have orbit gaps and therefore cannot serve as correct finite structural classifiers.

Corollary 6.17[93] (No Universal Exact-Certification Characterization Escapes the Binary Pairwise Obstruction). *Let* G *be a tractability predicate defined on all rigorously specified exact-correctness problems. Suppose that, for every binary pairwise slice* U*, the verdict of* G *on the canonical optimizer-set exact specification induced by* U *agrees with polynomial-time exact optimizer-set search on* U*. Then no exact characterization extracted from* G *decides the four obstruction families on the full binary pairwise domain. Equivalently, no universal exact-certification characterization over rigorously specified problems escapes the obstruction by passing to the semantic layer.*

Proof sketch. Corollary 2.38 places the canonical optimizer-set exact specifications of binary pairwise slices inside the universal exact-semantics framework. Restrict G to that witness class and apply the exact-specification no-go for optimizer-set search semantics. The contradiction is therefore inherited by any treatment claiming universal exact-certification scope. ■

Proposition 6.18[94] (Orbit-Gap Completeness for Exact Classification). *For any slice predicate* Q*, the following are equivalent:*

1. *there exists a closure-law-invariant predicate* P *such that* $P(S) \iff Q(S)$ *for every slice* S*;*
2. Q *is itself closure-law invariant;*
3. Q *is constant on closure orbits.*

Equivalently, exact classification of Q *by closure-law-invariant predicates fails if and only if* Q *admits an orbit-gap witness: two closure-equivalent slices with different* Q*-status.*

Proof sketch. The implication (1) ⇒ (2) transfers closure-law invariance across exact equivalence. The implication (2) ⇒ (3) says exactly that a closure-law-invariant predicate cannot change value inside a closure orbit. For (3) ⇒ (2), every primitive closure-law step is already a closure equivalence, so orbit constancy supplies the six invariance clauses. The final equivalence is the contrapositive of orbit constancy. ■

Orbit gaps are the complete obstruction criterion for exact classification of any fixed target predicate by closure-law-invariant classifiers. The four obstruction families are not claimed to exhaust all hard-side phenomena; they show that the same orbit-gap mechanism defeats several natural candidate frontier predicates.

Informally: only what matters can matter to a correct classifier.

[92] : `FR350` [93] : `FR351`, `FR361` [94] : `FR201-204`

Corollary 6.19[95] (Orbit-Gap Completeness on Closure-Closed Domains). *Let D be a closure-closed domain of slices, and let T be a target predicate on D. Then T admits an exact characterization on D by a closure-law-invariant predicate if and only if T has no orbit-gap witness inside D. Equivalently, exact characterization on D by closure-law-invariant predicates fails if and only if there exist $U, V \in D$ in the same closure orbit with different T-status.*

Proof sketch. The forward implication is immediate from closure-law invariance. For the reverse implication, take the closure hull of $D \cap T$; if T has no orbit-gap witness inside D, that closure hull agrees with T on D. ∎

Orbit Algebra of Exact Classification

The positive side of orbit-gap completeness is as important as the obstruction. Exact classification does exist when the target predicate is already constant on closure orbits, and the canonical classifier is obtained by orbit saturation.

Write

$$\mathrm{Hull}(Q)(U) \;:\Longleftrightarrow\; \exists V \left(V \sim_{\mathrm{cl}} U \wedge Q(V)\right)$$

for the closure hull of a slice predicate Q, where $\sim_{\mathrm{cl}}$ denotes closure equivalence.

Proposition 6.20[96] (Closure-Invariant Predicates Are Exactly the Fixed Points of Orbit Saturation). *For any slice predicate Q, the following are equivalent:*

1. *Q is closure-law invariant;*
2. *for every slice U,* $\mathrm{Hull}(Q)(U) \iff Q(U)$.

Equivalently, closure-law-invariant predicates are exactly the fixed points of the orbit-saturation operator Hull.

Proof sketch. If Q is closure-law invariant, it is constant on closure orbits, so adding all closure-equivalent slices does not enlarge it. Conversely, if Q agrees with its orbit saturation, then any closure-equivalent pair lies simultaneously inside or outside Q, which is exactly closure-law invariance. ∎

Theorem 6.21[97] (Exact Classification Equals Hull Separation). *Let D be a closure-closed domain and let Q be a target predicate on D. Then Q admits an exact characterization on D by a closure-law-invariant predicate if and only if the positive and negative orbit saturations are disjoint:*

$$\mathrm{Hull}(D \cap Q) \cap \mathrm{Hull}(D \cap \neg Q) = \varnothing.$$

Equivalently, exact classification fails exactly when one closure orbit meets both the positive and negative parts of the domain.

Proof sketch. The orbit-gap criterion says exact classification fails exactly when some closure orbit contains both a positive and a negative point of D. That is precisely the statement that the positive and negative orbit saturations overlap. Disjointness is therefore equivalent to exact classifiability. ∎

Corollary 6.22[98] (Least Exact Closure-Invariant Classifier). *Let D be a closure-closed domain and let Q have no orbit gaps on D. Then* $\mathrm{Hull}(D \cap Q)$ *is the least closure-law-invariant predicate that classifies Q exactly on D: it is correct on D, and every other exact closure-law-invariant classifier on D contains it.*

[95] : FR210-213 [96] : FR355 [97] : FR356-357 [98] : FR358

Proof sketch. Correctness on D follows from the no-orbit-gap assumption. Minimality follows because every exact closure-law-invariant classifier must contain the orbit saturation of the positive part $D \cap Q$. ■

Applied to exact tractability, Theorem 2.20 places every correct tractability classifier on a closure-closed domain inside this regime. Orbit gaps are therefore the complete obstruction criterion for exact closure-invariant classification, while the finite structural no-go concerns which of those classifiers remain accessible inside the local structural regime.

Corollary 6.23[99] (Domain Restriction Helps Only by Removing Orbit Gaps). *Let D be a closure-closed domain, and let Q be a target predicate on D such that correctness of a classifier for Q forces closure-orbit agreement on D. Then D admits a correct classifier for Q if and only if Q has no orbit-gap witness inside D. Equivalently, restricting the domain avoids the no-go only by eliminating all orbit gaps of Q on that restricted domain.*

Proof sketch. The reverse implication is Corollary 6.19. For the forward implication, correctness-forced orbit agreement rules out any same-orbit disagreement inside D. The artifact proves this abstract principle directly, and also formalizes the optimizer-computation instance as a concrete compute-cost theorem. ■

Domain restriction is governed by orbit-gap freedom on the restricted closure-closed domain. Excluding the four named witness families is not enough by itself. Those families witness orbit gaps, but any remaining orbit gap triggers the same impossibility theorem. If no orbit gap remains, the closure-hull construction yields a correct classifier on that domain.

7 Related Work

Rough Sets, Feature Selection, and Exact Relevance

At the static level, rough-set reduct theory [16, 26] is the clearest classical comparison point. That literature studies which attributes can be deleted while preserving decision distinctions. Exact static sufficiency is a reduct condition for the induced decision table. The question here is different: which tractable mechanisms admit a finite characterization?

A separate literature studies feature selection, variable importance, and model explanation in machine learning and AI, including wrapper and filter methods [12], general surveys [10], and attribution methods based on Shapley-style decompositions [13]. These works matter here by motivation rather than by technique. They typically optimize predictive performance, explanatory salience, or approximation quality under data-dependent criteria. Exact relevance certification instead asks for a semantic preservation property: which coordinates are necessary to preserve the outputs that a correctness condition counts as admissible. Section 2 proves that arbitrary exact admissible-output semantics reduce to this same quotient-recovery problem. This includes decision, counting, search, approximation guarantees, PAC/regret/risk guarantees, finite-horizon or anytime guarantees, randomized-output guarantees, and distributional specifications written as admissible-output relations.

Decision-theoretic informativeness and value of information, beginning with Blackwell's comparison of experiments and subsequent decision-theoretic treatments [3, 19, 11], provide a second comparison point. Both settings ask when one representation retains all decision-relevant distinctions of another. The difference here is the combinatorial complexity of certifying exact preservation under coordinate deletion.

[99] : `FR252-256`

Information-Theoretic and Transport Viewpoints

The zero-distortion discussion belongs to the classical information-theoretic tradition of exact distinguishability and lossless coding boundaries, going back to Shannon and standard modern treatments of source coding and rate-distortion theory [21, 22, 23, 5]. Fisher-style support counting supplies another classical comparison lens on relevant support [7]. These viewpoints are used only structurally. There is no general coding or statistical estimation problem here. Zero distortion, entropy, and support counting appear only to show that several independent summary frameworks are already constrained by the same optimizer-quotient core. The Wasserstein language in Section 4 plays the same role: a qualitative description of quotient branching rather than a full transport-theoretic model.

Backdoors, Tractable Islands, and Dichotomy Programs

Backdoor tractability for constraint satisfaction is the nearest complexity-theoretic analogue [27, 2, 8]. There, a small structural set exposes membership in a tractable class even when the ambient problem is hard. Exact relevance certification likewise asks which coordinates matter, and polynomial-time behavior emerges when the source of hardness is restricted by structure. Bessiere et al. [2] make especially clear that exploiting a known tractable structure and discovering the responsible structure are different algorithmic tasks; later parameterized work sharpened that distinction. The same separation holds between positive tractable mechanisms and the harder meta-level problem of recognizing a correct frontier classifier. The bounded-actions, bounded-treewidth, bounded-support, and related positive cases can also be read through the parameterized-complexity lens as tractable parameter regimes, even though no single parameter is claimed to capture the full frontier.

The dichotomy tradition for satisfiability and constraint satisfaction provides the broader analogy, from Schaefer's Boolean dichotomy theorem [20] to the finite-domain CSP dichotomy theorems of Bulatov and Zhuk [4, 28]. The expectation that a clean tractability boundary should exist was articulated in the Feder–Vardi program [6], and later algebraic work of Barto and Kozik clarified why finite structural boundaries are plausible in that setting [1]. The analogy is narrower and methodological rather than classificatory: for output specifications of any kind, quotient realizability and closure-law invariance defeat the most direct admissible classifier, first at the feature-sufficiency layer and then, by representative instantiations, at compute-cost tasks such as optimizer computation and explicit admissibility-preserving output search. Any successful frontier theorem must therefore use stronger structure than the direct closure-invariant regime of Definition 2.19. No complete analogue of polymorphisms is identified; the result isolates the regime that is already ruled out.

The quotient viewpoint also touches the literature on exact abstraction and aggregation in stochastic control, Markov decision processes, and reinforcement learning [15, 9, 17]. Those works study when states may be aggregated while preserving value or policy structure. Our setting is narrower: we ask for exact preservation of optimizer classes under coordinate-hiding maps rather than arbitrary state abstractions. Optimizer equivalence is therefore closer to a reward- or optimizer-relevant abstraction than to full bisimulation. Still, the common theme is that semantic invariants of decision behavior should survive presentation-level simplification.

Meta-Impossibility Traditions

The closest analogy is a Rice-style impossibility theorem on a specific representation domain. Rice's theorem [18] is unconditional: nontrivial extensional properties of partial recursive functions are undecidable, with extensionality forced by what "semantic" means rather than chosen as an axiom. Refinements such as the Myhill–Shepherdson theorem and the Rice–Shapiro theorem extend this pattern to broader structured semantic domains [14, 24]. The present theorem has the same invariance-plus-disagreement shape with a different forced invariance and a narrower scope. Proposition 2.1 shows that exact certification depends only on quotient data, and Theorem 2.20 lifts this to tractability classification on closure-closed domains: any correct tractability classifier, for any output specification written as an admissible-output relation, must agree on closure-equivalent representations. The no-go then depends on this forced invariance alone (Theorem 6.11), with Corollary 6.14 giving the resulting impossibility for the direct structural regime studied here. The remaining admissibility clauses play the role of algorithmic and structural guardrails that exclude classifiers admissible in name only, but they do not enter the contradiction.

8 Conclusion

Any rigorously specified computational problem determines a state-indexed admissible-output relation R, the only state distinctions that matter are the admissible-output equivalence classes $s \sim_R s' \iff \mathrm{Adm}_R(s) = \mathrm{Adm}_R(s')$, and every exact correctness claim reduces to the same quotient-recovery problem. Exact relevance certification asks which coordinates recover those classes. Section 2 proves that exact predicates, decision problems, counting, search, approximation, PAC/regret/risk, finite-horizon or anytime, randomized-output, and distributional guarantees all reduce to that same quotient-recovery problem. Every state equivalence relation is realizable at this semantic layer, every exact claim admits a coordinate presentation, rigorously specified problems lie in the exact-certification framework itself, and semantically extensional claims factor through the same quotient. Exact correctness is not structure imposed by formal methods; it is the structure a specification must carry for objective correctness, and hence tractability of the underlying problem, to be well-posed at all. Representative compute-cost instantiations include optimizer computation, canonical payload and search tasks, and external or transported outputs with explicit admissibility-preserving transport; correctness of a polynomial-time classifier for any of these tasks forces the same closure-orbit agreement. The same realizability results show that the framework is universal representationally as well as semantically: arbitrary labeling kernels already arise as optimizer quotients.

On the positive side, the tractable landscape factors into primitive mechanisms, lifts, and degeneracies. The eight-mechanism basis is the explicit inventory of the current positive landscape, not a complete classification of all tractability mechanisms for exact relevance certification. Proposition 2.26 shows that this positive side remains substantive at the representation level: bounded distinct action profiles compress to bounded-action slices without changing exact certification, so the existing bounded-actions theorem already yields a genuine structural classifier on that subcase. The frontier side now has its own positive theory as well: closure-invariant predicates are exactly the fixed points of orbit saturation, exact classifiability is exactly hull disjointness, and when a target is exactly classifiable there is a least exact closure-invariant classifier. On the negative side, quotient realizability is maximal, closure-law invariance is forced by correctness, and the orbit-gap mechanism defeats finite structural classifiers across the four named obstruction predicates on the full binary pairwise slice domain. Corollary 6.17 makes the scope explicit: no universal exact-certification characterization over rigorously specified problems escapes this obstruction. The binary pairwise witness class already sits inside the universal semantic framework through canonical optimizer-set exact specifications, so universality expands the reach of the no-go rather than providing a way around it. The same pair-targeted action-independent affine transport yields both the finite-structural collapse theorem and the concrete compute-cost no-go for polynomial-time optimizer computation. Domain restriction changes the theorem only by removing orbit gaps on the restricted closure-closed domain.

Simple finite lists, unconstrained quotient predicates, and closure-invariant classifiers of this finite structural kind cannot characterize the boundary. Any successful frontier theorem must use stronger representation-sensitive structure than the one ruled out here.

Successor Structural Classes

The CSP dichotomy program gives the clearest methodological comparison: successful frontier theorems use algebraic or comparably global invariants that survive quotient semantics without collapsing to the direct closure-invariant structural regime ruled out here. A concrete candidate with existing proof support is the offset-normalized decision-relevant interaction graph on coordinate-symmetric binary pairwise slices: Proposition 6.3 shows that symmetry collapses it to an empty-versus-complete dichotomy, so any sharper frontier inside that regime has to refine a canonical two-point algebra rather than arbitrary sparse graph patterns. The natural next candidate beyond that symmetry subregime is a polymorphism-style algebraic structure on the optimizer map itself, because the CSP analogy suggests that tractability should first be sought in operations preserving the semantic object rather than in presentation-local patterns. If that route is too coarse, the next natural level is the sufficient-set family or minimal-sufficient lattice, where one can ask for quotient-invariant global constraints stronger than bounded local syntax. A third possibility is to restrict the transformation group itself, replacing the full affine closure by a stricter class of admissible transports and asking whether a frontier theorem survives on that smaller orbit structure. More broadly, one can look for other transport-compatible global constraints that survive closure orbits while still ruling out arbitrary quotient geometry. No such complete invariant is identified. Uniform strict-gap control preserves the full optimizer quotient, while arbitrarily small perturbations can still flip relevance and sufficiency. In applied settings, that instability is the practical headline: without explicit gap control, approximation alone does not license claims about which coordinates matter. Exactness therefore remains the reference semantics against which approximate relevance claims have to be validated.

Declaration of Generative AI and AI-assisted technologies in the writing process

AI tools were used for prose refinement, notation, and LaTeX editing, and for drafting candidate formalizations. The author reviewed and edited the output under human supervision and retained full intellectual and editorial control. Candidate formalizations were then checked in Lean 4; no load-bearing technical claim was accepted without Lean verification and direct author review. The author is solely responsible for all statements, citations, and conclusions.

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